

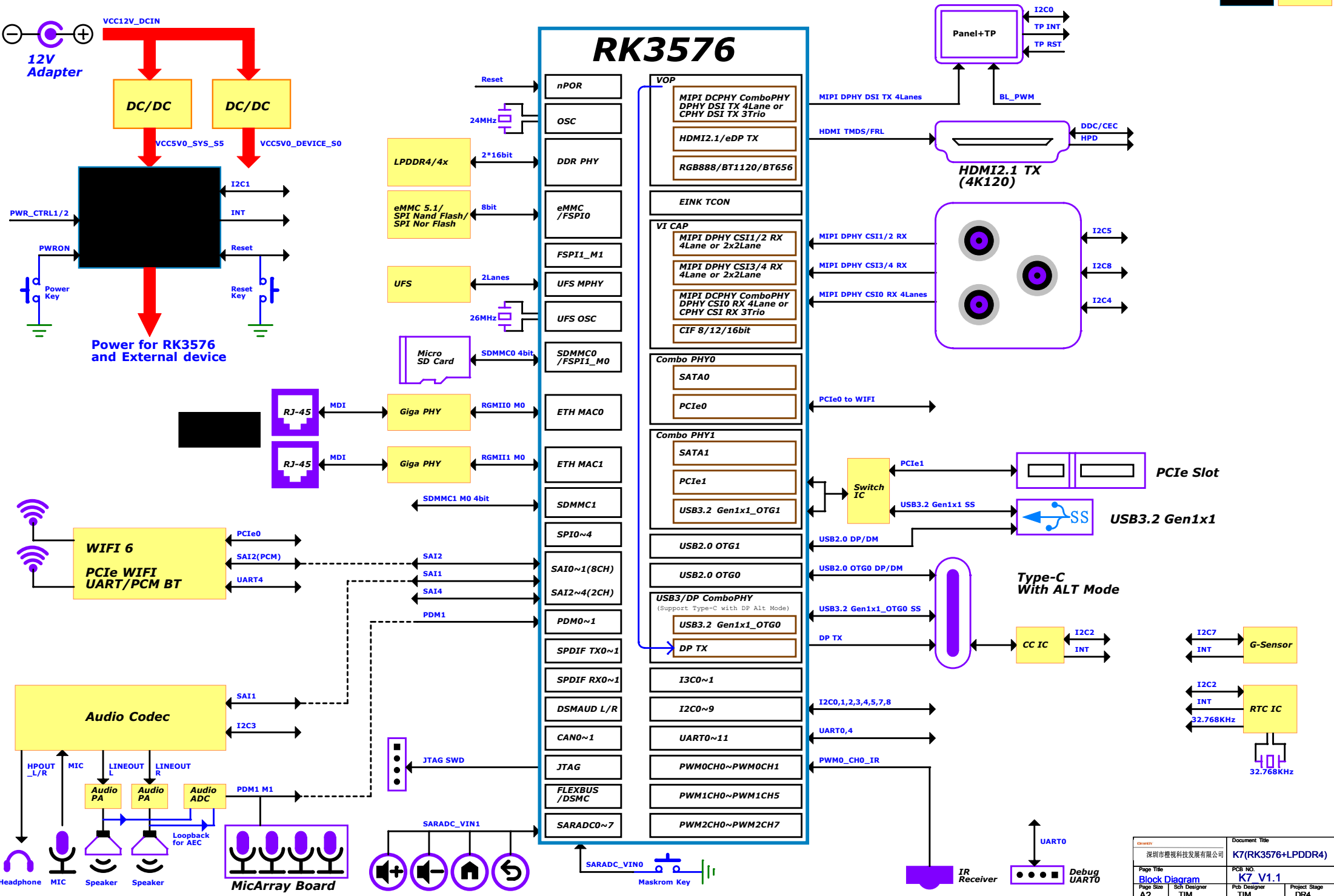
Note:

The RK806S-5 LDO power distribution of the reference schematic is only suitable for the interface used in the reference schematic.
If other interface functions need to be added to the reference schematic, the RK806S-5 LDO distribution needs to be re evaluated, otherwise the added functions may exceed the maximum current provided by the LDO

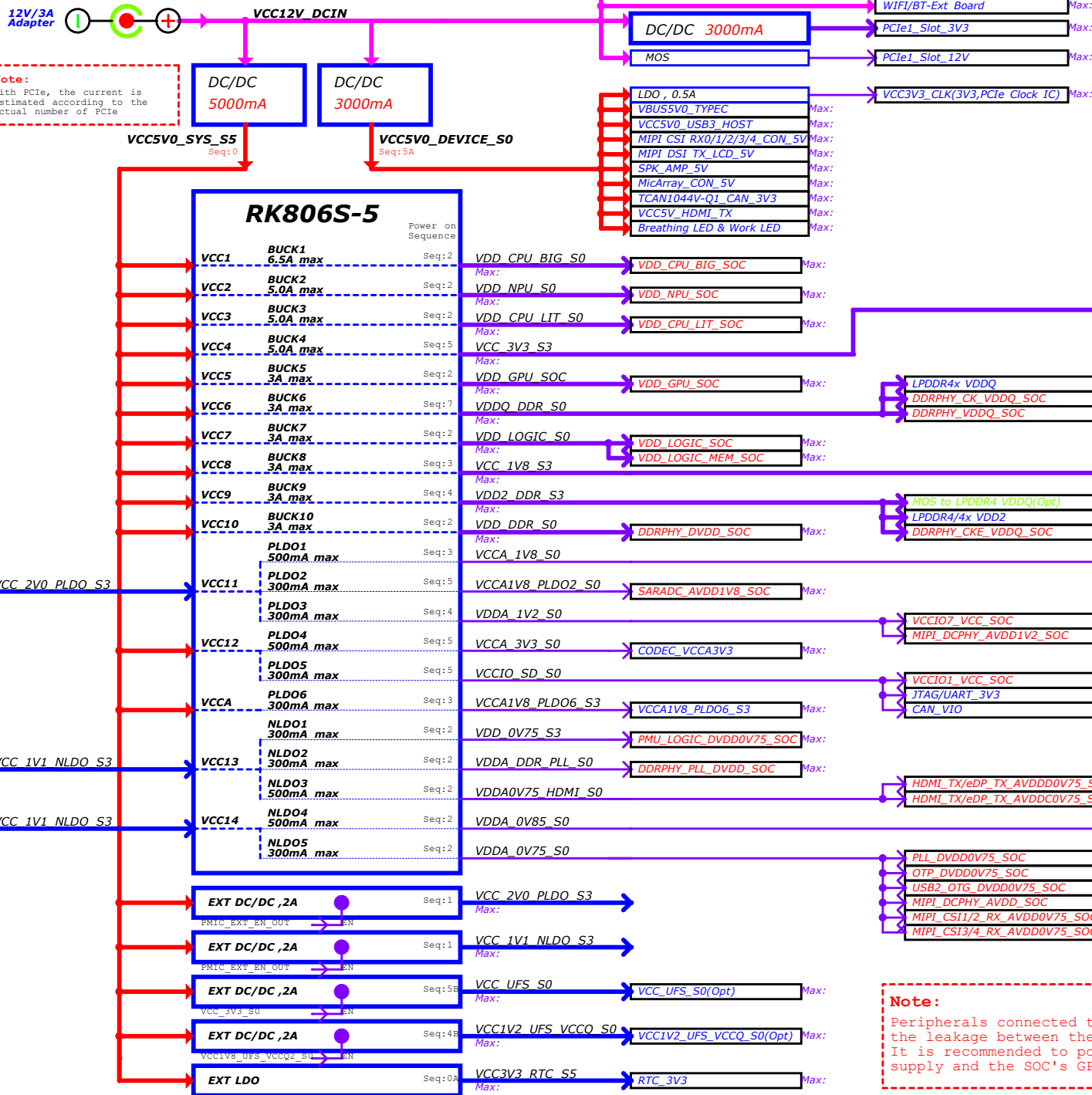
Dranth 深圳市橙视科技发展有限公司		Document Title K7(RK3576+LPDDR4)	
Page Title Cover Page		PCB NO. K7_V1.1	
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RK3576 Ref Block Diagram(Typical Application Case)

Other IC



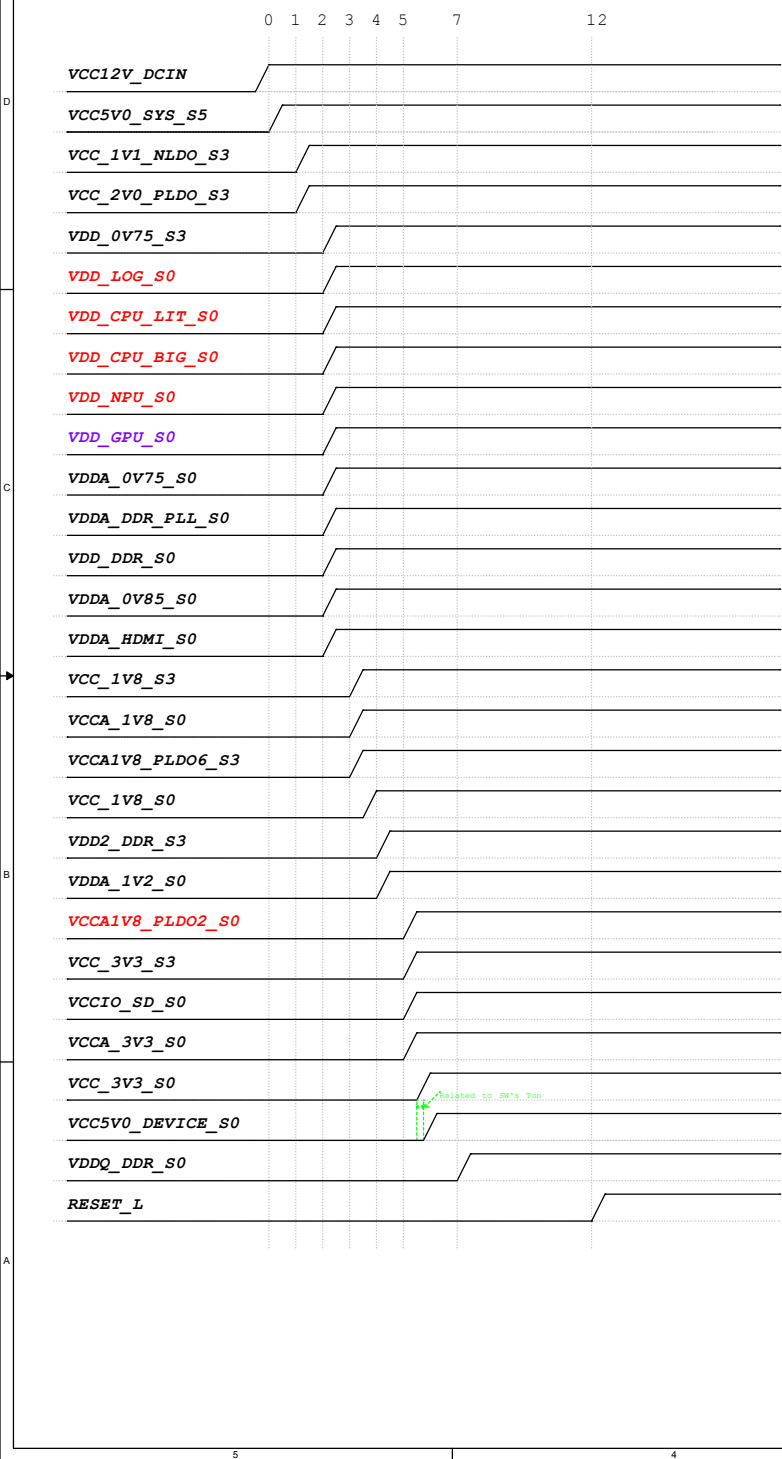
Power Tree



Note:
The RK806S-5 LDO power distribution of the reference schematic is only suitable for the interface used in the reference schematic. If other interface functions need to be added to the reference schematic, the RK806S-5 LDO distribution needs to be re evaluated, otherwise the added functions may exceed the maximum current provided by the LDO

Note:
Peripherals connected to the GPIO of SOC need to consider the leakage between the GPIO of SOC and the Peripherals. It is recommended to power on both the Peripherals's power supply and the SOC's GPIO power supply simultaneously.

Power Sequence



Power description

Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Work Voltage	Peak Current	Sleep Current
VCC5V0_SYS_S5	RK806_BUCK1	6.5A	VDD_CPU_BIG_S0	Slot:2	0.85V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK2	5A	VDD_NPU_S0	Slot:2	0.75V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK3	5A	VDD_CPU_LIT_S0	Slot:2	0.85V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK4	5A	VCC_3V3_S3	Slot:5	3.3V	ON	3.3V	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK5	3A	VDD_GPU_S0	Slot:2	ADJ FB=0.5V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK6	3A	VDDQ_DDR_S0	Slot:7	ADJ FB=0.5V	ON	0.61V-LP4/4x 0.51V-LP5	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK7	3A	VDD_LOGIC_S0 VDD_LOGIC_MEM_S0	Slot:2	0.75V	ON	0.75V	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK8	3A	VCC_1V8_S3	Slot:3	1.8V	ON	1.8V	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK9	3A	VDD2_DDR_S3	Slot:4	ADJ FB=0.5V	ON	1.1V-LP4/4x 1.05V-LP5	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK10	3A	VDD_DDR_S0	Slot:2	0.85V	ON	0.85V DVFS	TBD	TBD
VCC_2V0_PLDO	RK806_PLDO1	0.5A	VCCA_1V8_S0	Slot:3	1.8V	ON	1.8V	TBD	TBD
	RK806_PLDO2	0.3A	VCCA1V8_PLDO2_S0	Slot:5	1.8V	ON	1.8V	TBD	TBD
	RK806_PLDO3	0.3A	VDDA_1V2_S0	Slot:4	1.2V	ON	1.2V	TBD	TBD
VCC5V0_SYS_S5	RK806_PLDO4	0.5A	VCCA_3V3_S0	Slot:5	3.0V	ON	3.3V	TBD	TBD
	RK806_PLDO5	0.3A	VCCIO_SD_S0	Slot:5	3.3V	ON	3.3V	TBD	TBD
VCC5V0_SYS_S5	RK806_PLDO6	0.3A	VCCA1V8_PLDO6_S3	Slot:3	1.8V	ON	1.8V	TBD	TBD
VCC_1V1_NLDO	RK806_NLDO1	0.3A	VDD_0V75_S3	Slot:2	0.75V	ON	0.75V	TBD	TBD
	RK806_NLDO2	0.3A	VDDA_DDR_PLL_S0	Slot:2	0.85V	ON	0.85V DVFS	TBD	TBD
	RK806_NLDO3	0.5A	VDDA0V75_HDMI_S0	Slot:2	0.75V	ON	0.75V	TBD	TBD
VCC_1V1_NLDO	RK806_NLDO4	0.5A	VDDA_0V85_S0	Slot:2	0.85V	ON	0.85V	TBD	TBD
	RK806_NLDO5	0.3A	VDDA_0V75_S0	Slot:2	0.75V	ON	0.75V	TBD	TBD
	RK806_RESETh								
VCC5V0_SYS_S5	EXT BUCK	2A	VCC_2V0_PLDO_S3	Slot:1	2.1V	ON	2.0V	TBD	TBD
VCC5V0_SYS_S5	EXT BUCK	2A	VCC_1V1_NLDO_S3	Slot:1	1.1V	ON	1.1V	TBD	TBD
VCC12V_DCIN	EXT BUCK	5A	VCC5V0_SYS_S5	Slot:0	5.0V	ON	5.0V	TBD	TBD
VCC12V_DCIN	EXT BUCK	3A	VCC5V0_DEVICE_S0	Slot:5A	5.2V	ON	5.2V	TBD	TBD
VCC_3V3_S3	SWITCH	2A	VCC_3V3_S0	Slot:5A	3.3V	ON	3.3V	TBD	TBD
VCC_1V8_S3	SWITCH	2A	VCC_1V8_S0	Slot:3A	1.8V	ON	1.8V	TBD	TBD

Note:

The power suffix S0, S3 or S5 means:
S5: Keep power on during power down
S3: Keep power on during sleeping
S0: Power off during sleeping

Note:

Peripherals connected to the GPIO of SOC need to consider the leakage between the GPIO of SOC and the Peripherals. It is recommended to power on both the Peripherals's power supply and the SOC's GPIO power supply simultaneously.

IO Power Domain Map

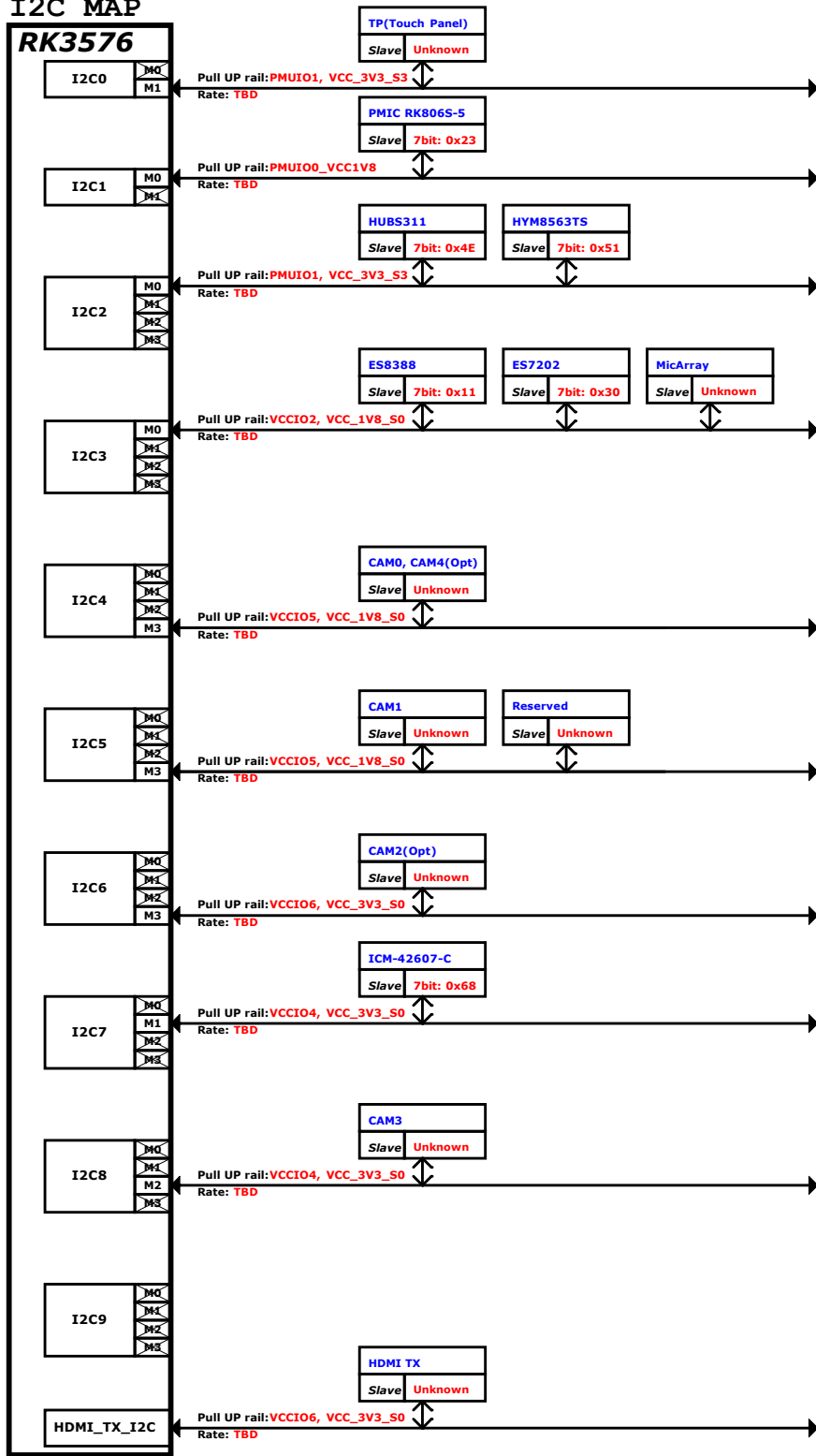
IO Domain	Pin Num	Support IO Voltage	Supply Power Pin Name	Power Source	Operating Voltage
PMUIO0	Pin 2K11	1.8V Only	PMUIO0_VCC1V8	VCC_1V8	1.8V
PMUIO1	Pin 1U20	1.8V or 3.3V	PMUIO1_VCC	VCC_1V8 VCC_3V3	3.3V
VCCIO0	Pin 1J20	1.8V Only	VCCIO0_VCC1V8	VCC_1V8	1.8V
VCCIO1	Pin 2A8	1.8V or 3.3V	VCCIO1_VCC	VCC_1V8 VCC_3V3	1.8V/3.3V
VCCIO2	Pin 2A2	1.8V or 3.3V	VCCIO2_VCC	VCC_1V8 VCC_3V3	1.8V
VCCIO3	Pin 2B10	1.8V or 3.3V	VCCIO3_VCC	VCC_1V8 VCC_3V3	1.8V
VCCIO4	Pin 2A7	1.8V or 3.3V	VCCIO4_VCC	VCC_1V8 VCC_3V3	3.3V
VCCIO5	Pin 2A4/2A5	1.8V or 3.3V	VCCIO5_VCC	VCC_1V8 VCC_3V3	1.8V
VCCIO6	Pin 2N3	1.8V or 3.3V	VCCIO6_VCC	VCC_1V8 VCC_3V3	3.3V
VCCIO7	Pin 2M3	1.2V or 1.8V	VCCIO7_VCC	VCC_1V2 VCC_1V8	1.2V

IO Type	Operating Voltage
1.8V Only	VCCIO*_VCC1V8=1.8V
1.2V or 1.8V	VCCIO*_VCC=1.2V or 1.8V
1.8V or 3.3V	VCCIO*_VCC=1.8V or 3.3V

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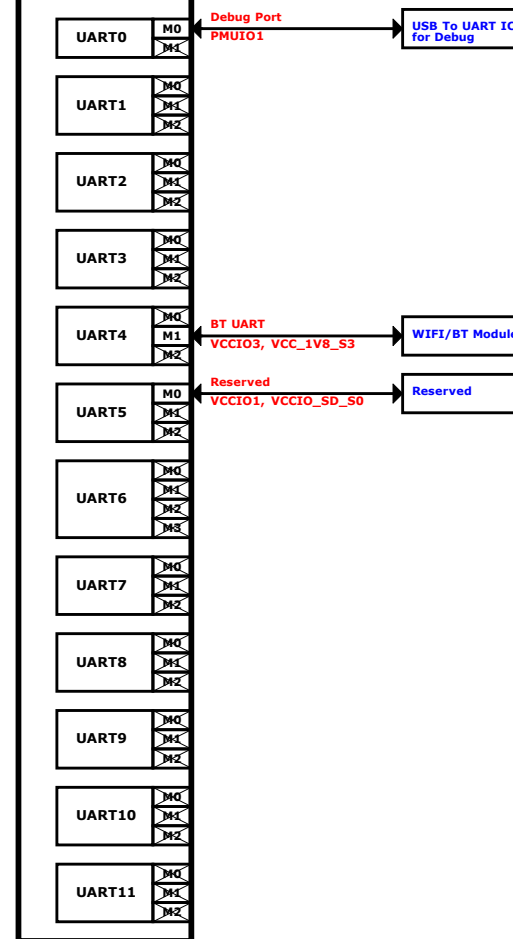
I2C MAP

RK3576



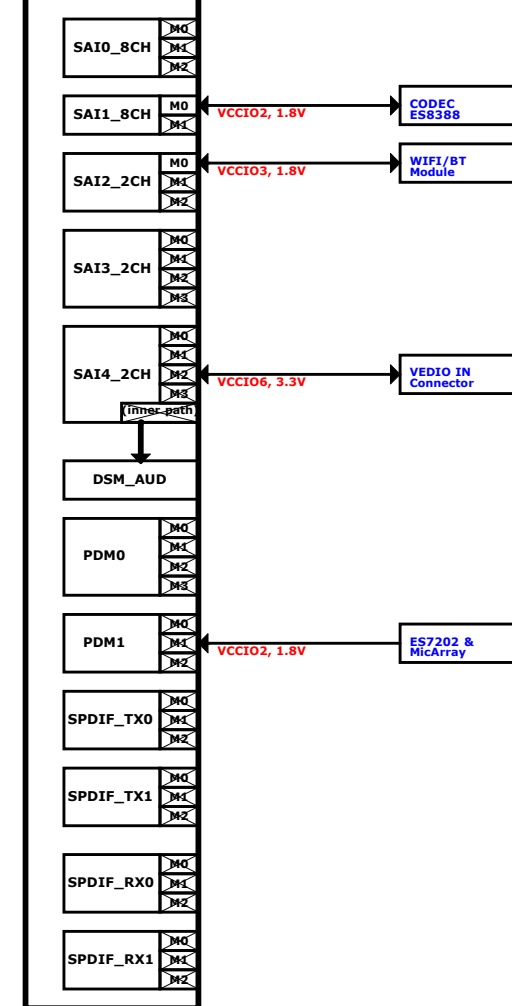
UART MAP

RK3576



Audio MAP

RK3576



Note:



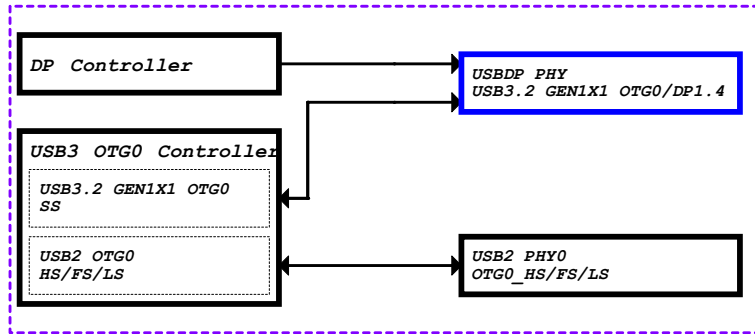
Unselected IOMUX path

IOMUX path in use

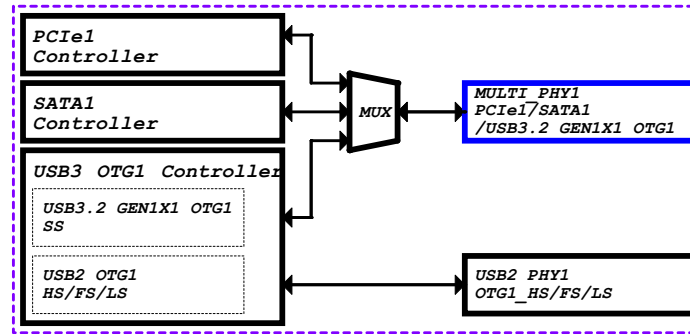
At the same time, only one path can be selected.

MULTI_PHY Path Map

USBDP PHY--USB3.2 GEN1X1 OTG0/DP1.4



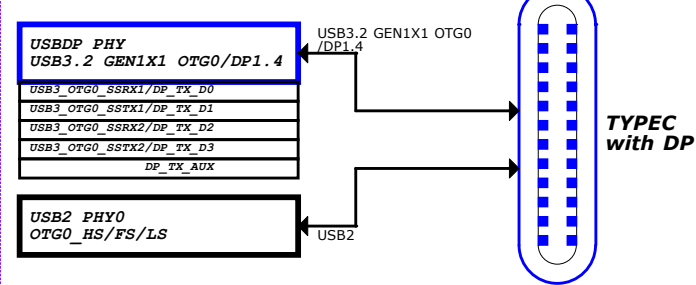
PCIe Combo PHY1-- PCIe1/SATA1/USB3.2 GEN1X1 OTG1



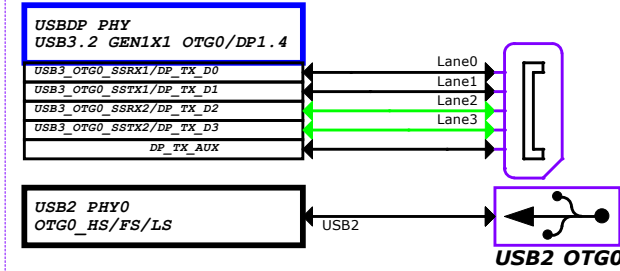
Note:
USB2 PHY1 can only be used when
PCIe1/SATA1 is not in use!!!

USB OTG0/DP Application

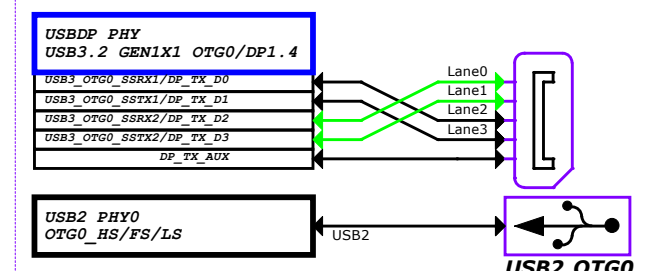
CASE0: TYPEC with DP



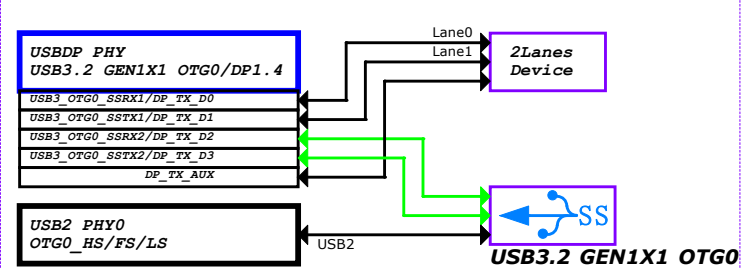
CASE1: USB2 OTG0 + DP 4Lane (Swap OFF)



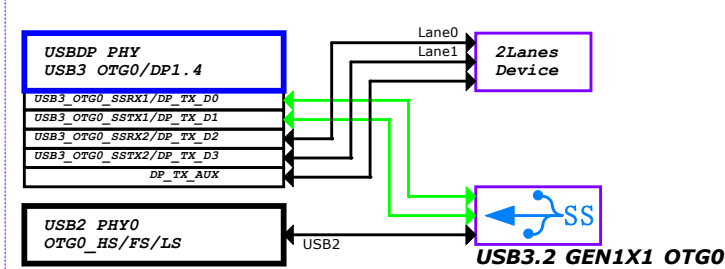
CASE2: USB2 OTG0 + DP 4Lane (Swap ON)



CASE3: USB3.2 GEN1X1 OTG0 + DP 2Lane (Swap OFF)



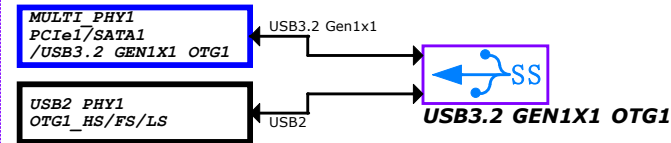
CASE4: USB3.2 GEN1X1 OTG0 + DP 2Lane (Swap ON)



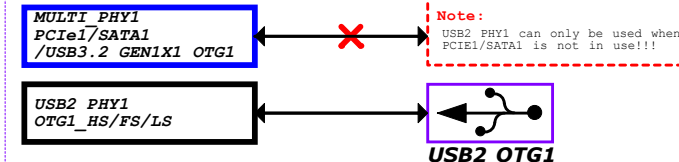
Note:
DP Lane swap enable
0: Lane0/1/2/3 TxData mapping to Lane0/1/2/3 TXDP/N
1: Lane0/1/2/3 TxData mapping to Lane2/3/0/1 TXDP/N

USB OTG1 Application

CASE0: USB3.2 GEN1X1 OTG1

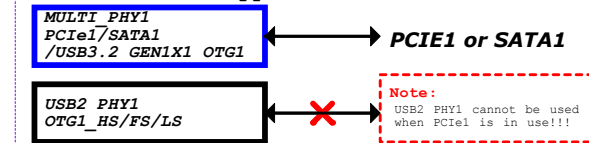


CASE1: USB2 OTG1

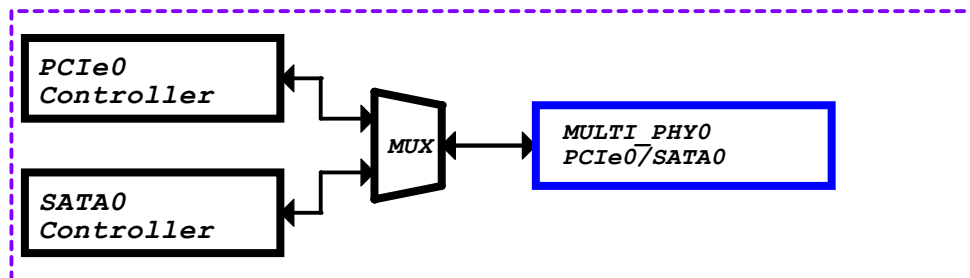


Note:
USB2 PHY1 can only be used when
PCIe1/SATA1 is not in use!!!

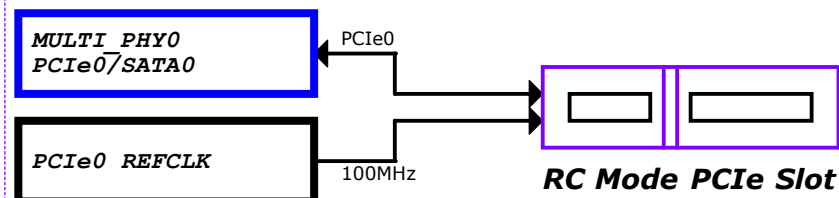
CASE2: Do not support USB



PCIE Combo PHY0--PCIE0/SATA0



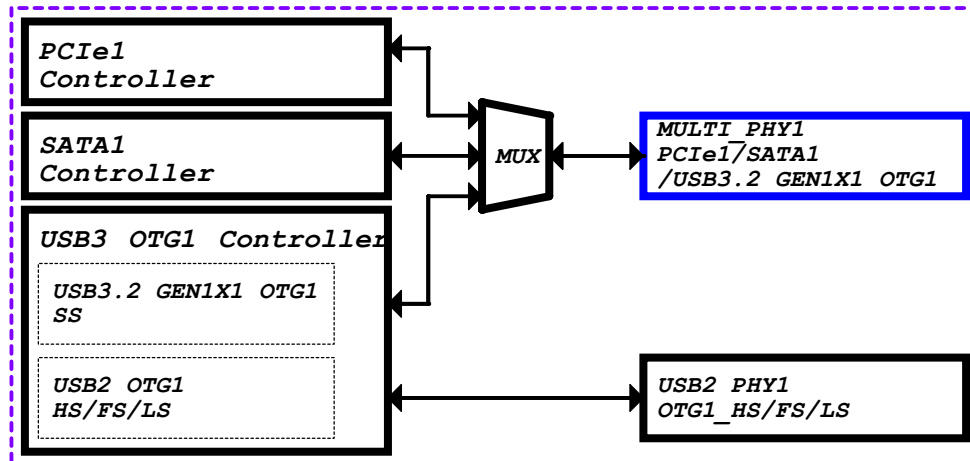
CASE0: PCIE x 1Lane



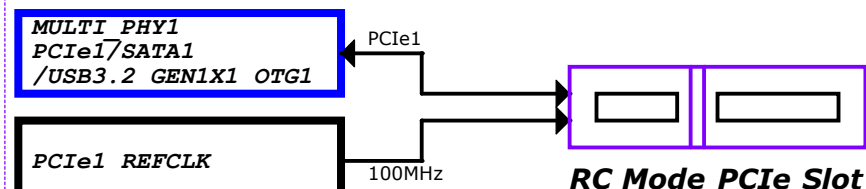
CASE1: SATA



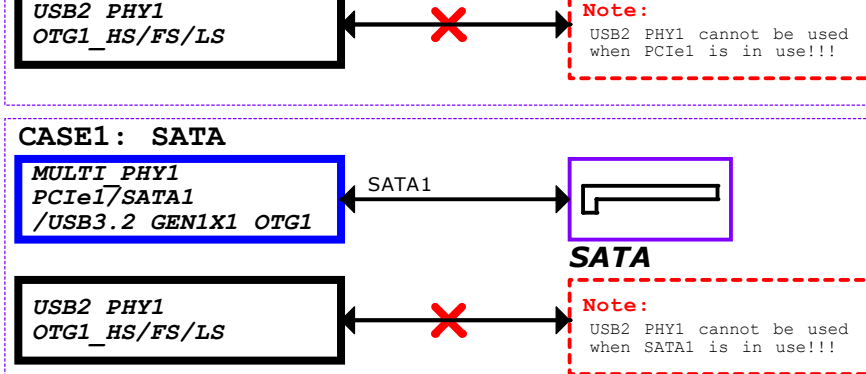
PCIE Combo PHY1--PCIE1/SATA1/USB3.2 GEN1X1 OTG1



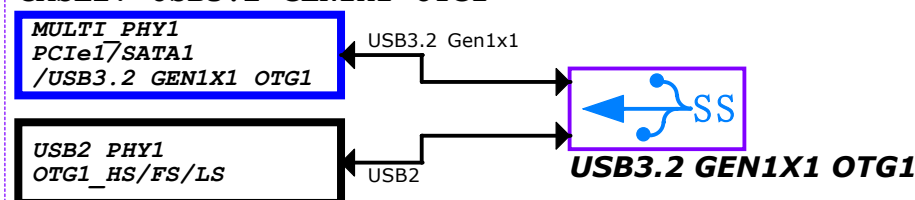
CASE0: PCIE x 1Lane



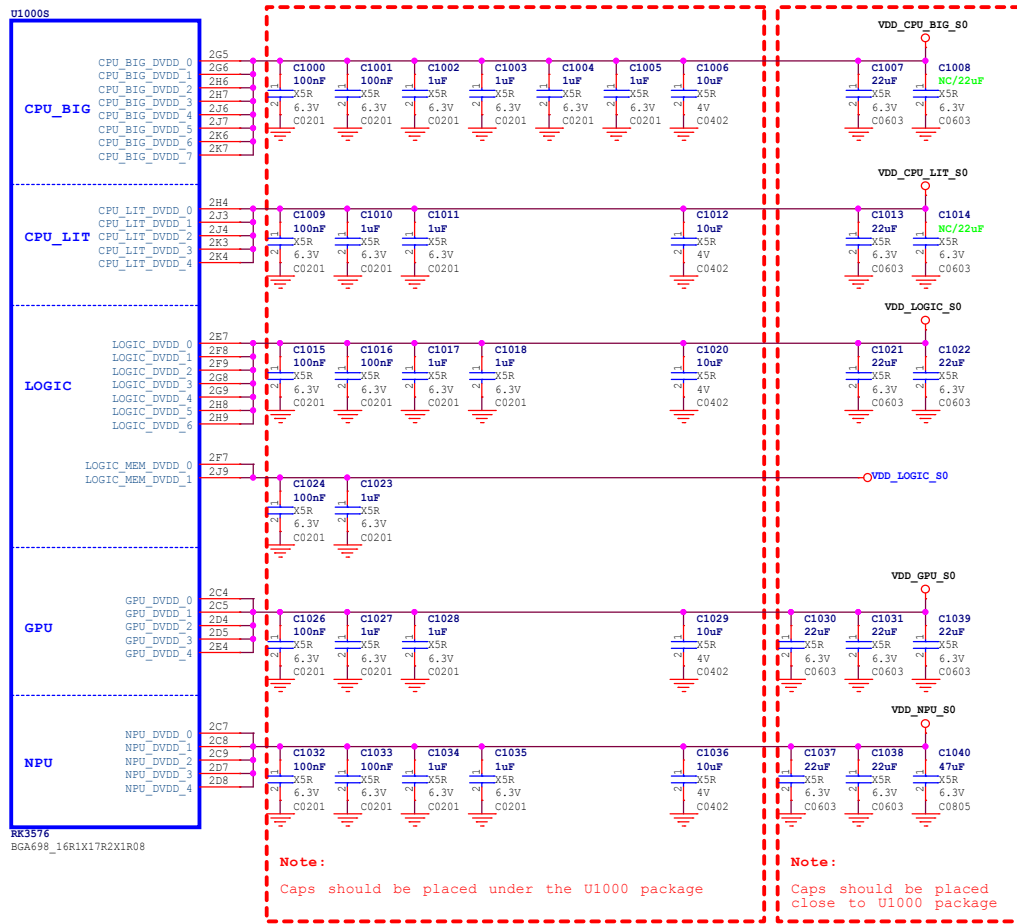
CASE1: SATA



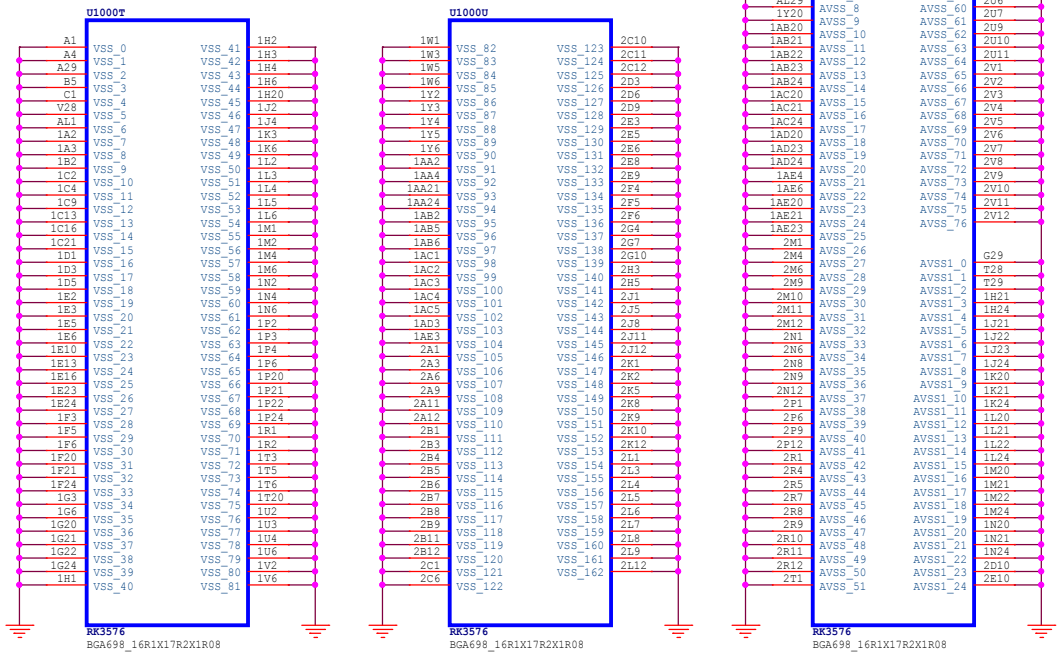
CASE2: USB3.2 GEN1X1 OTG1



RK3576 S (Power)

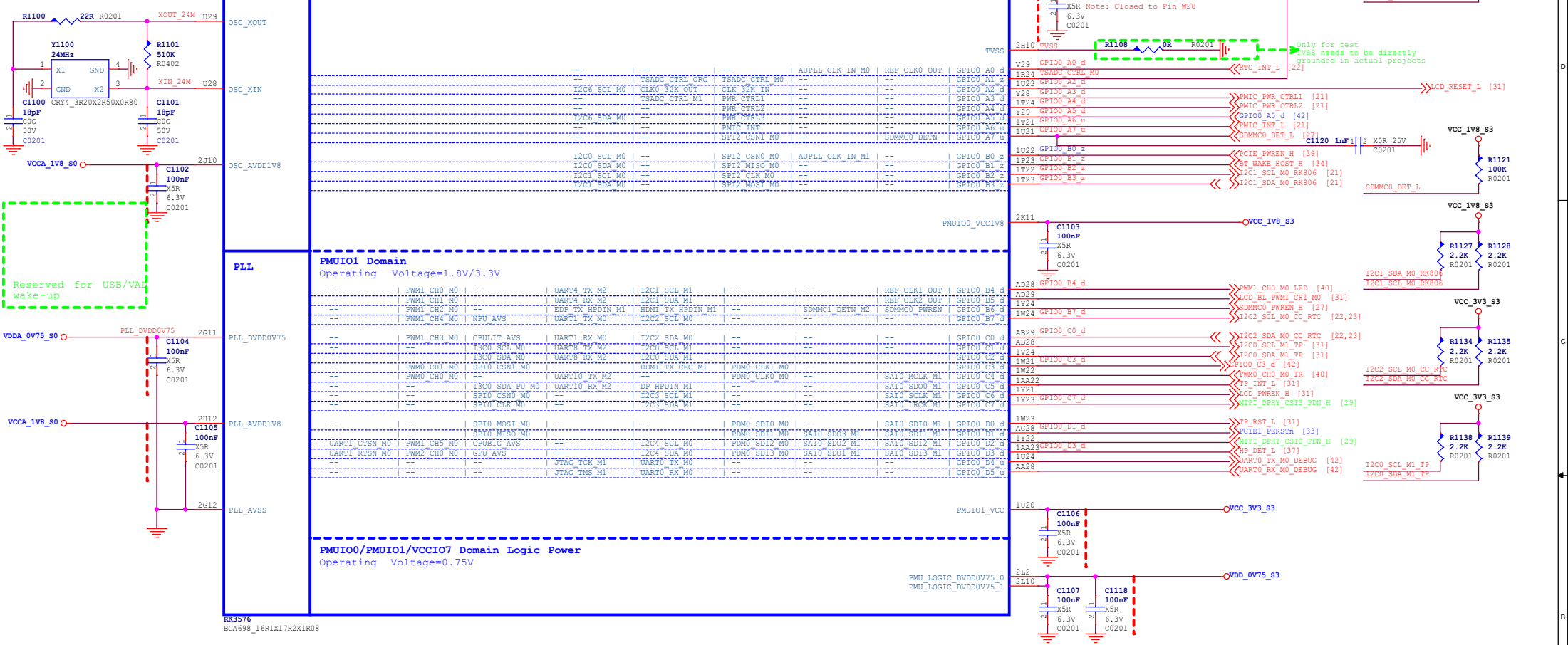


RK3576 T/U/V (GND)

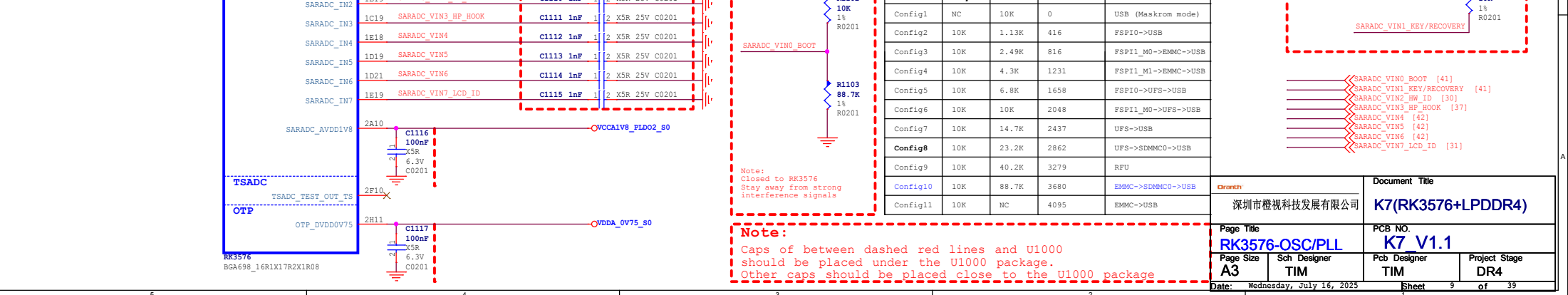


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RK3576 E
(PMUIO0/1)



RK3576 R
(SARADC)



U1000B

UFS2.0
 PWM-G1/G2/G3/G4
 HS-G1/G2/G3

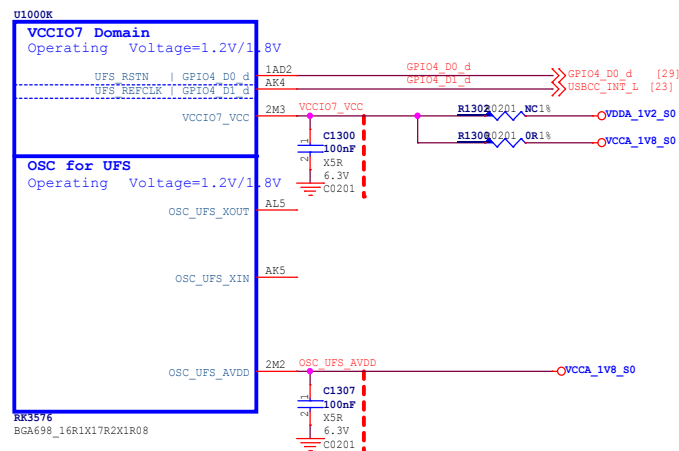
UFS_TX_D0P 1A06
 UFS_TX_D0N 1A06
 UFS_TX_D1P 1A04
 UFS_TX_D1N 1A05
 UFS_RX_D0P AK8
 UFS_RX_D0N AL7
 UFS_RX_D1P AK7
 UFS_RX_D1N AL6
 UFS_TX_REXT 2R2

UFS_AVDD0V85 2N2
 UFS_AVDD1V8 2P2

UFS_TX_REXT R1301 1% 8.2K R0201

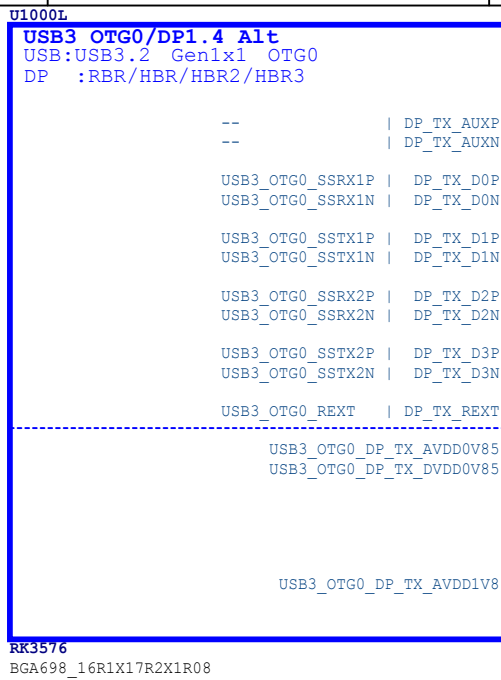
C1302 1uF X5R 6.3V C0201
 C1303 1uF X5R 6.3V C0201
 C1305 1uF X5R 6.3V C0201
 C1306 1uF X5R 6.3V C0201

BGA698
 16R1X17R2X1R0R

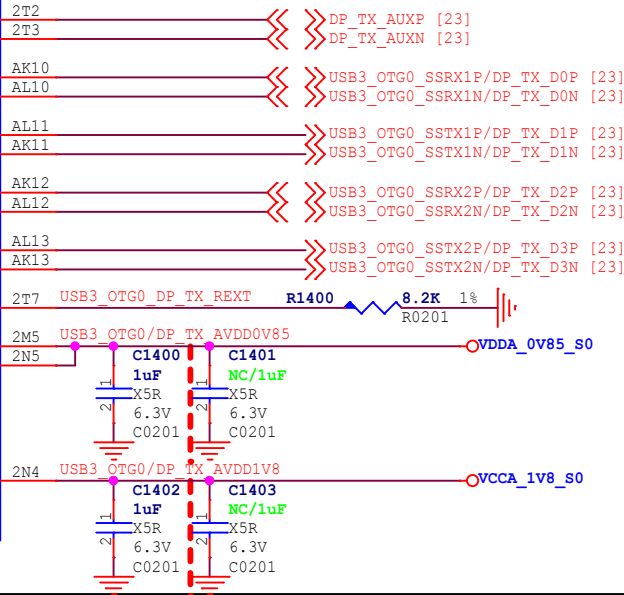
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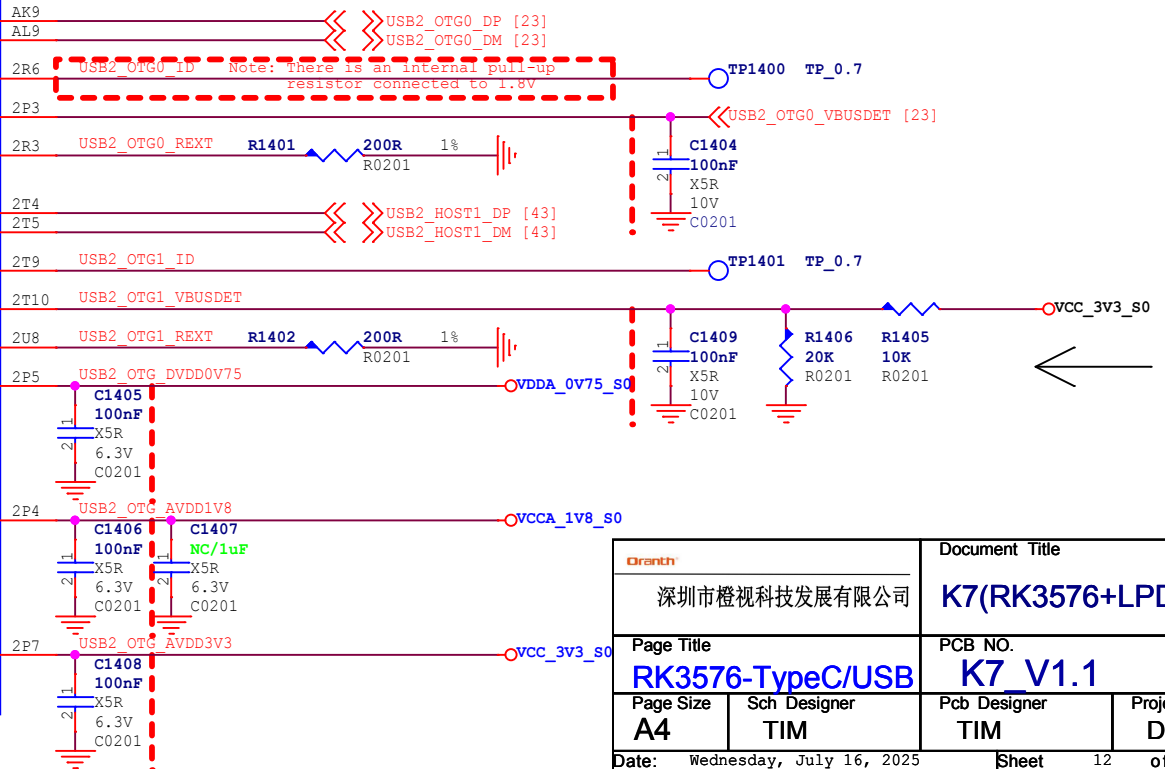
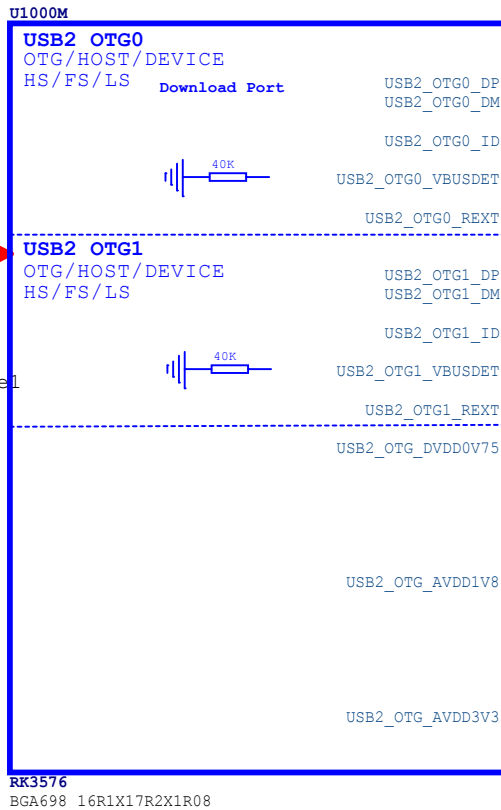
RK3576 L (USB3/DP)



Support:
Type-C With Displayport Alternate Mode



RK3576 M (USB2)

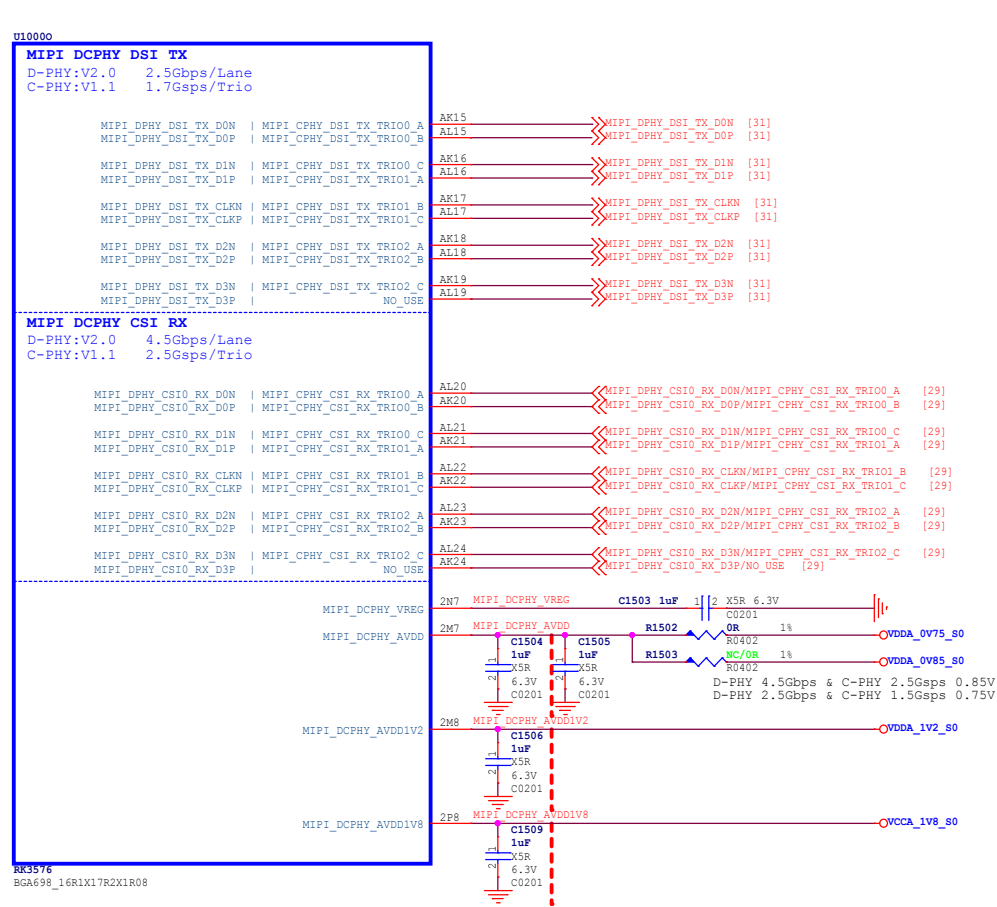


Note!!!

The USB2 PHY1 function cannot be used, if the PCIe1 or SATA1 function of Combo PHY1 is selected

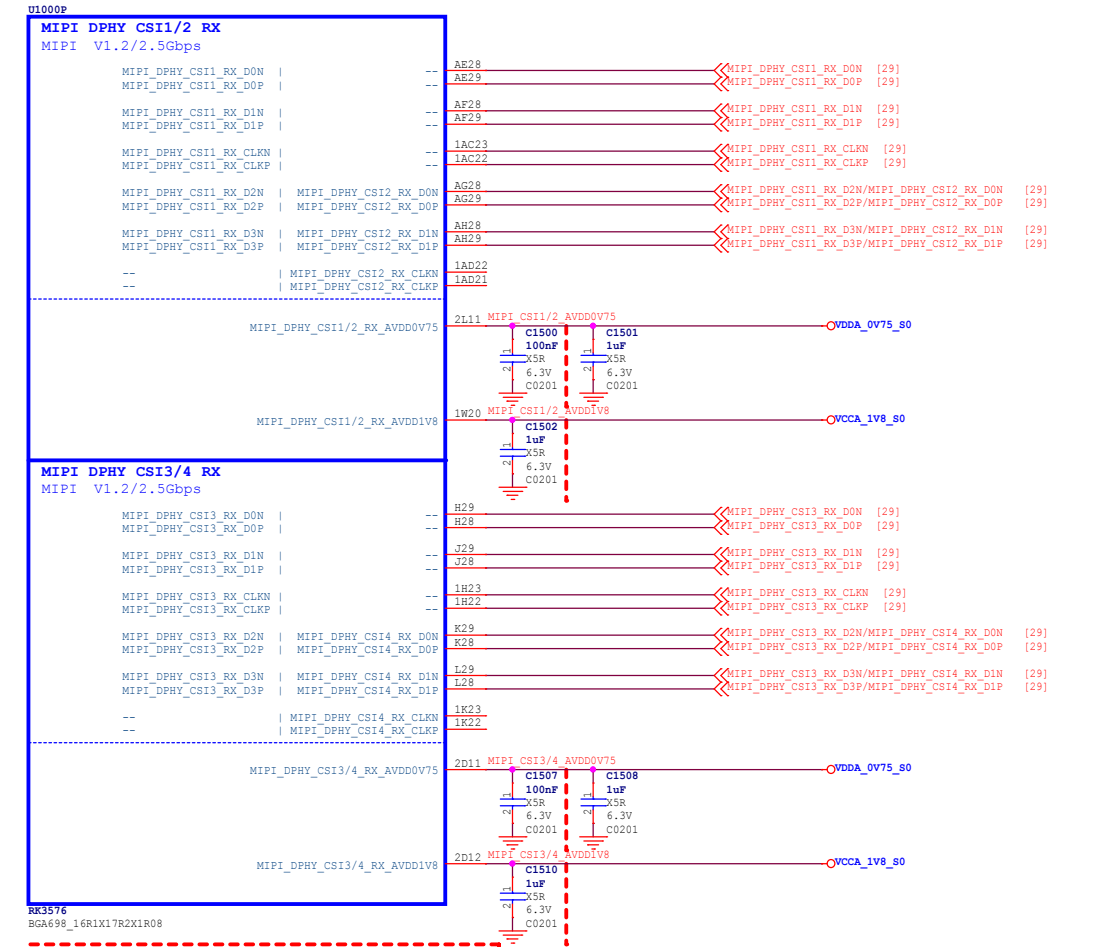
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RK3576_O (MIPI DCPHY)



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3576_P (MIPI DPHY CSI RX)

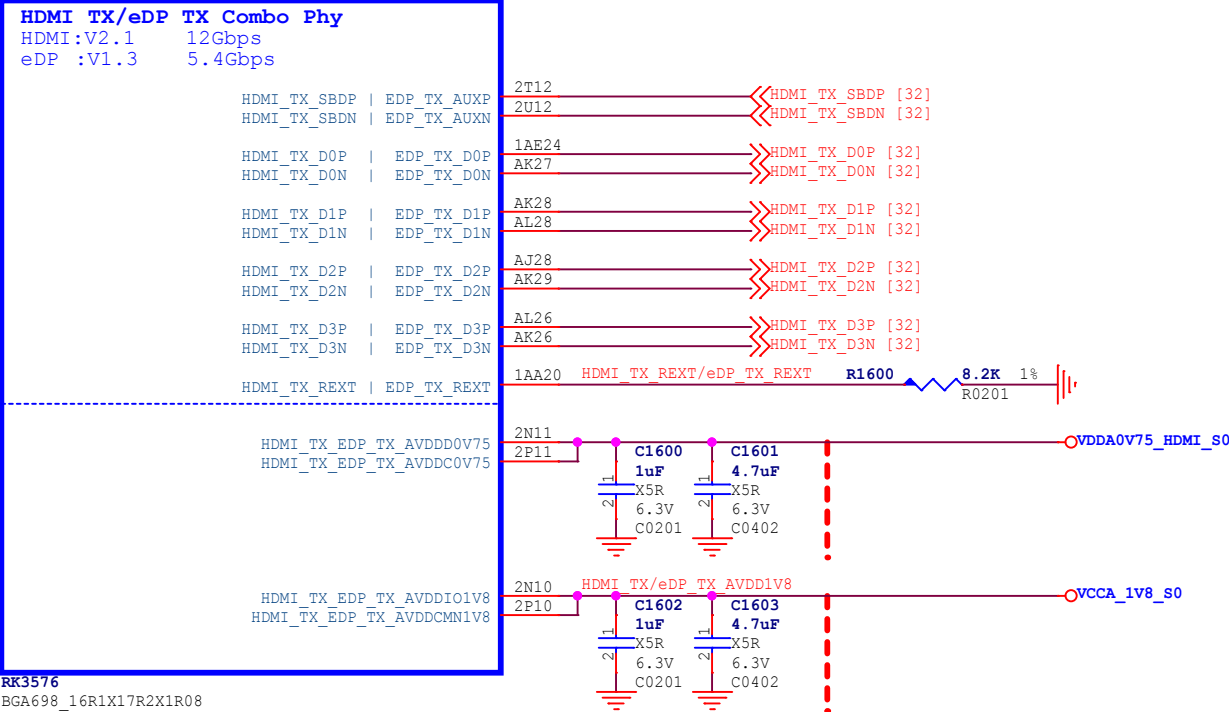


Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

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RK3576_Q (HDMI/eDP)

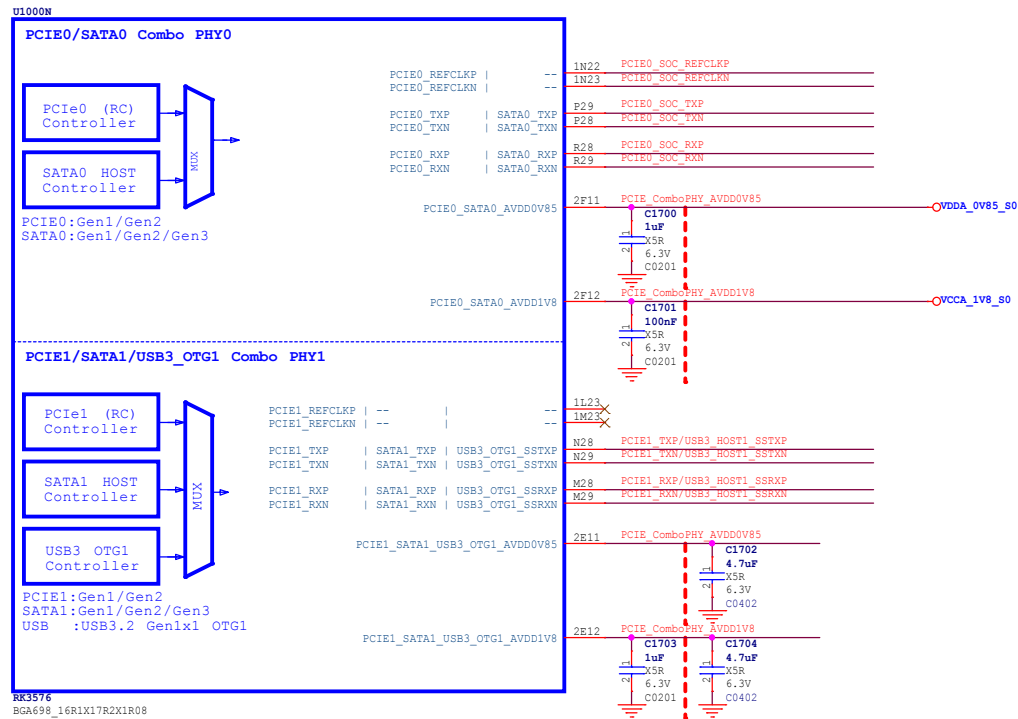
Note:
HDMI 2.1 supports up to 4Kx2K@120Hz
U1000Q



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

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RK3576_N (PCIe/SATA/USB3)

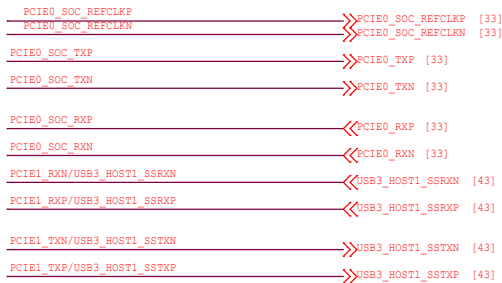


Note!!!

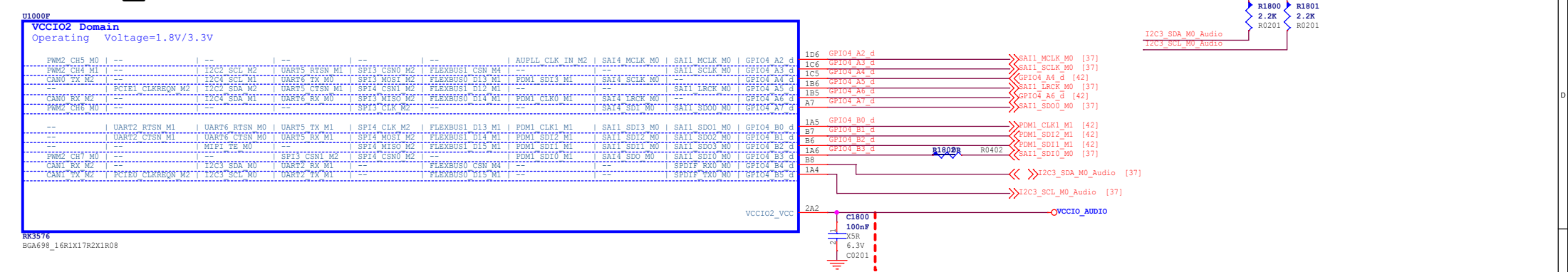
If the PCIe1 or SATA1 function of Combo PHY1 is selected, the USB3 OTG1 function cannot be used, and even the USB2 PHY1 function cannot be used

Note:

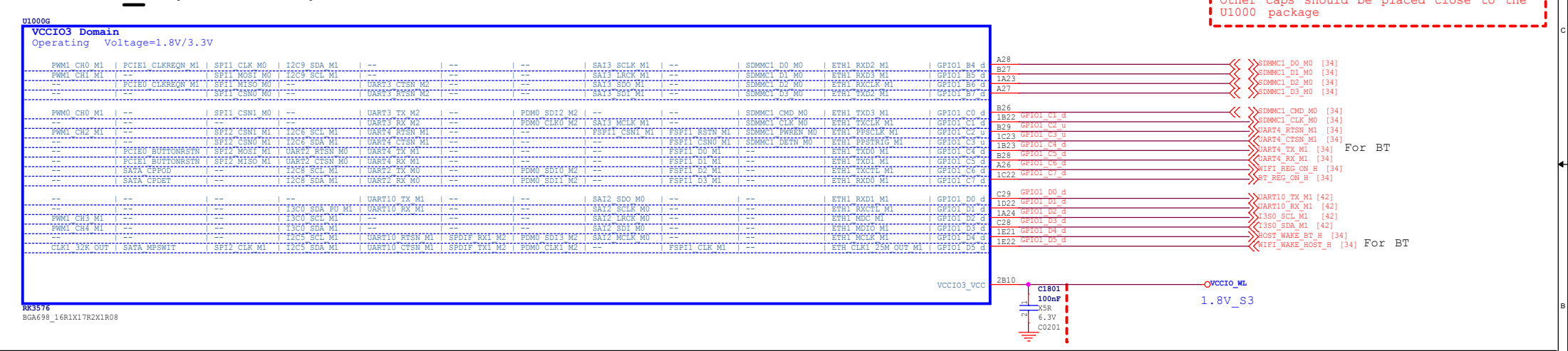
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package



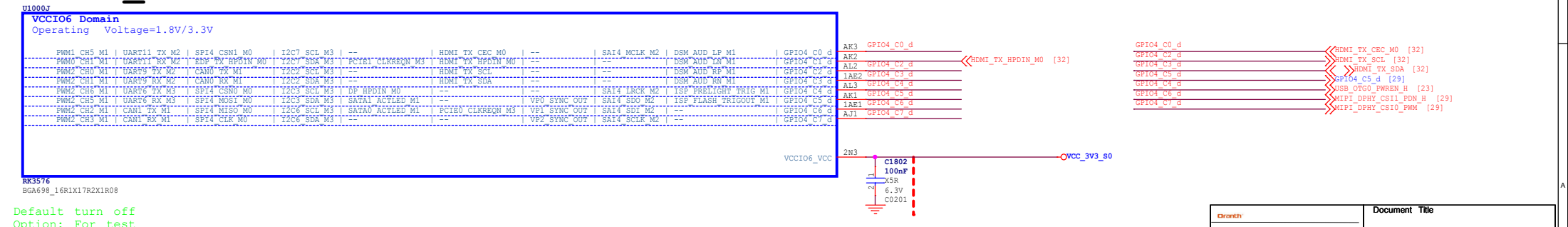
RK3576_F (VCCIO2)



RK3576 G (VCCIO3)



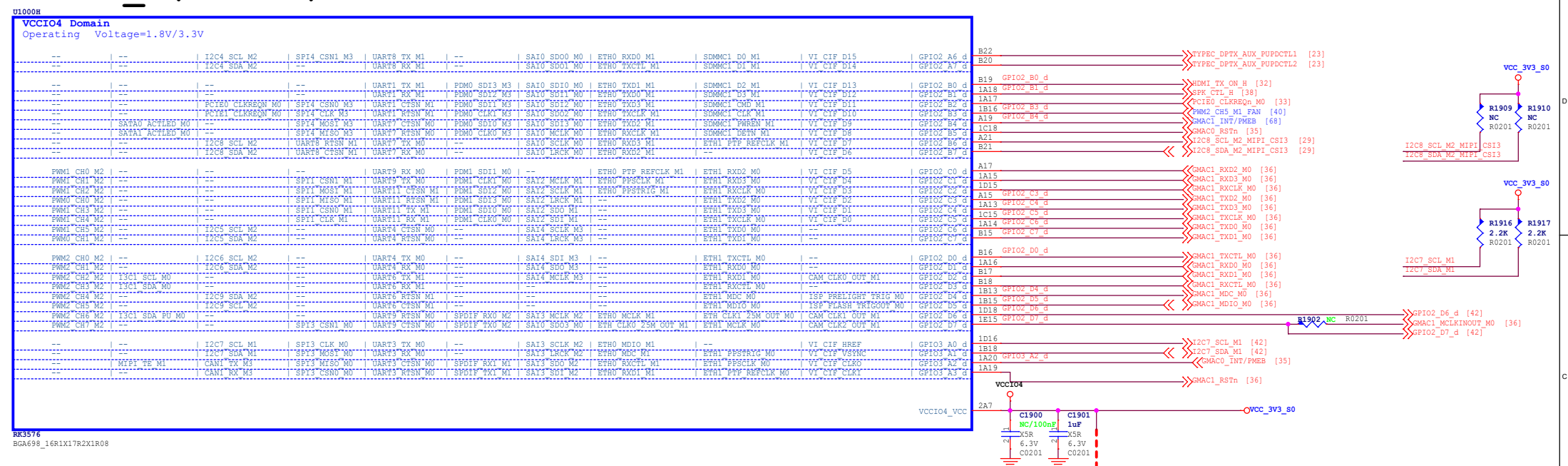
RK3576 J (VCCIO6)



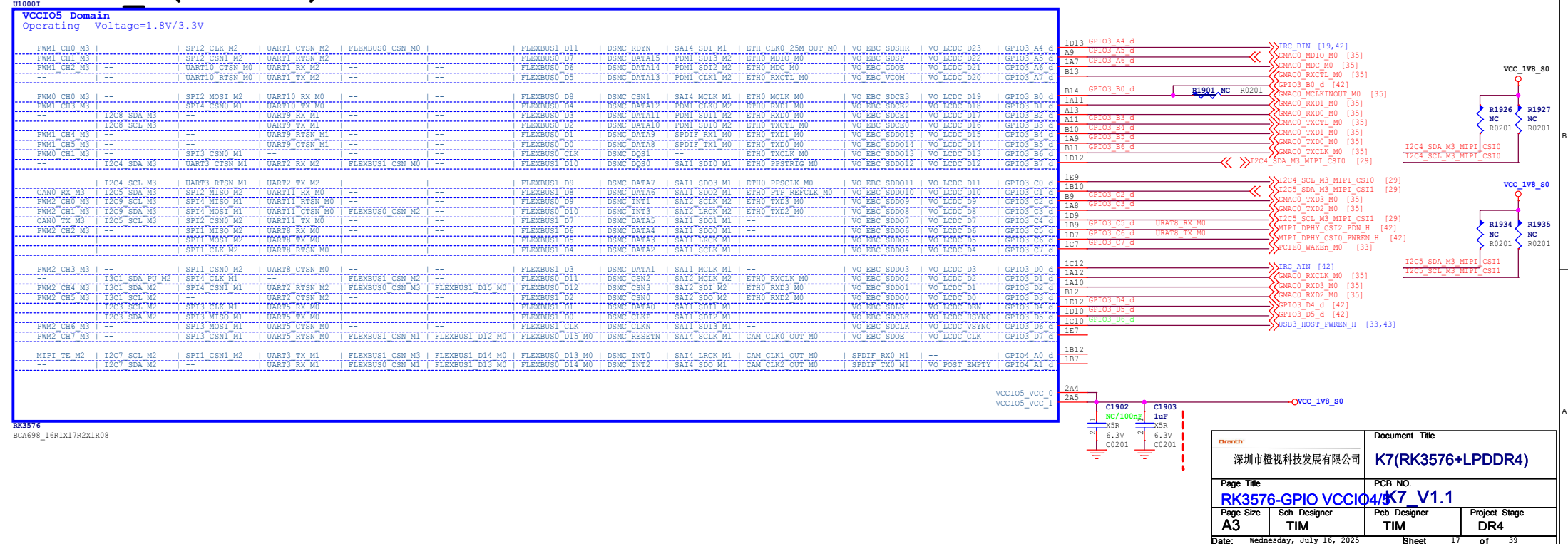
```
Default  turn off
Option:  For test
```

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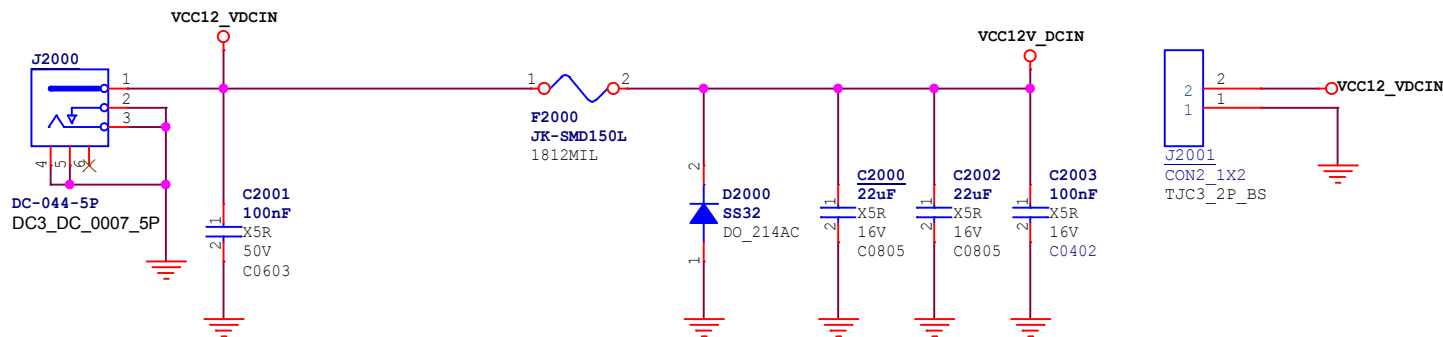
RK3576_H (VCCIO4)



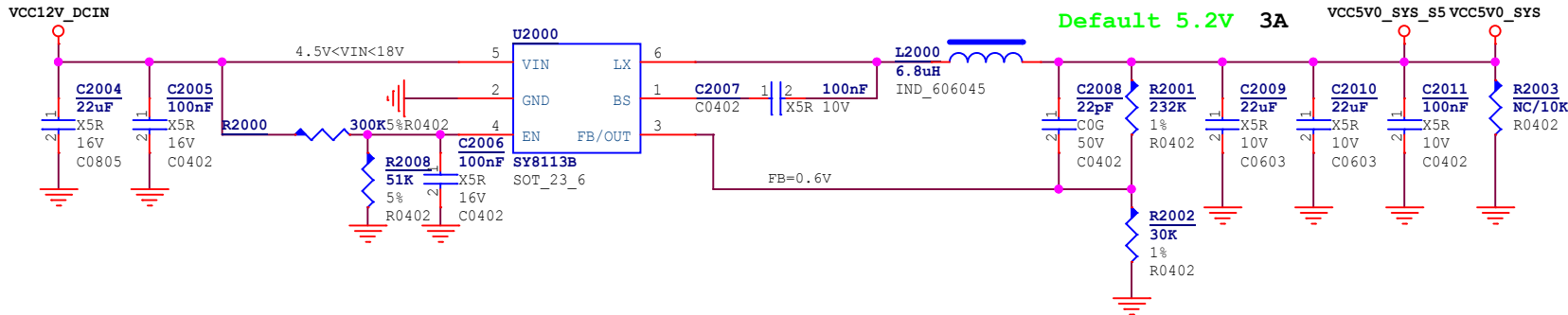
RK3576 I (VCCIO5)



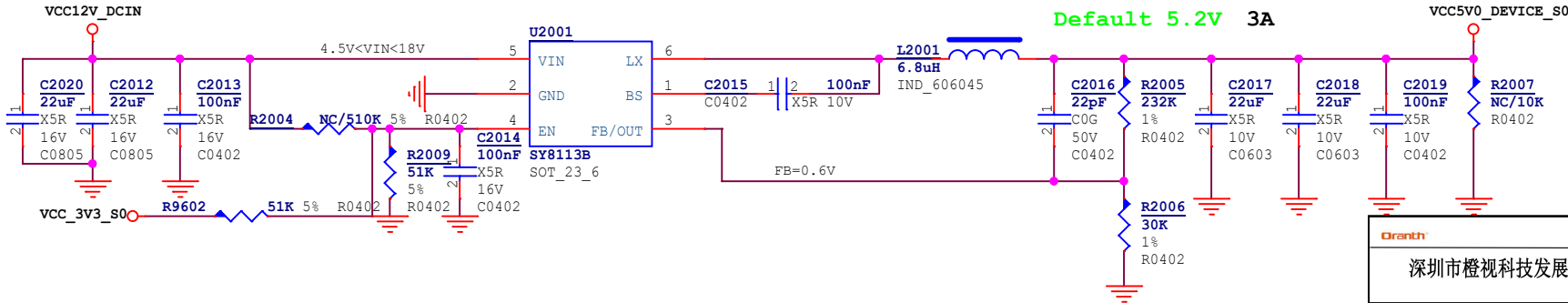
12V/3A DCIN



VCC5V0_SYS



VCC5V0_DEVICE



Doranth		Document Title	
深圳市橙视科技发展有限公司		K7(RK3576+LPDDR4)	
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```

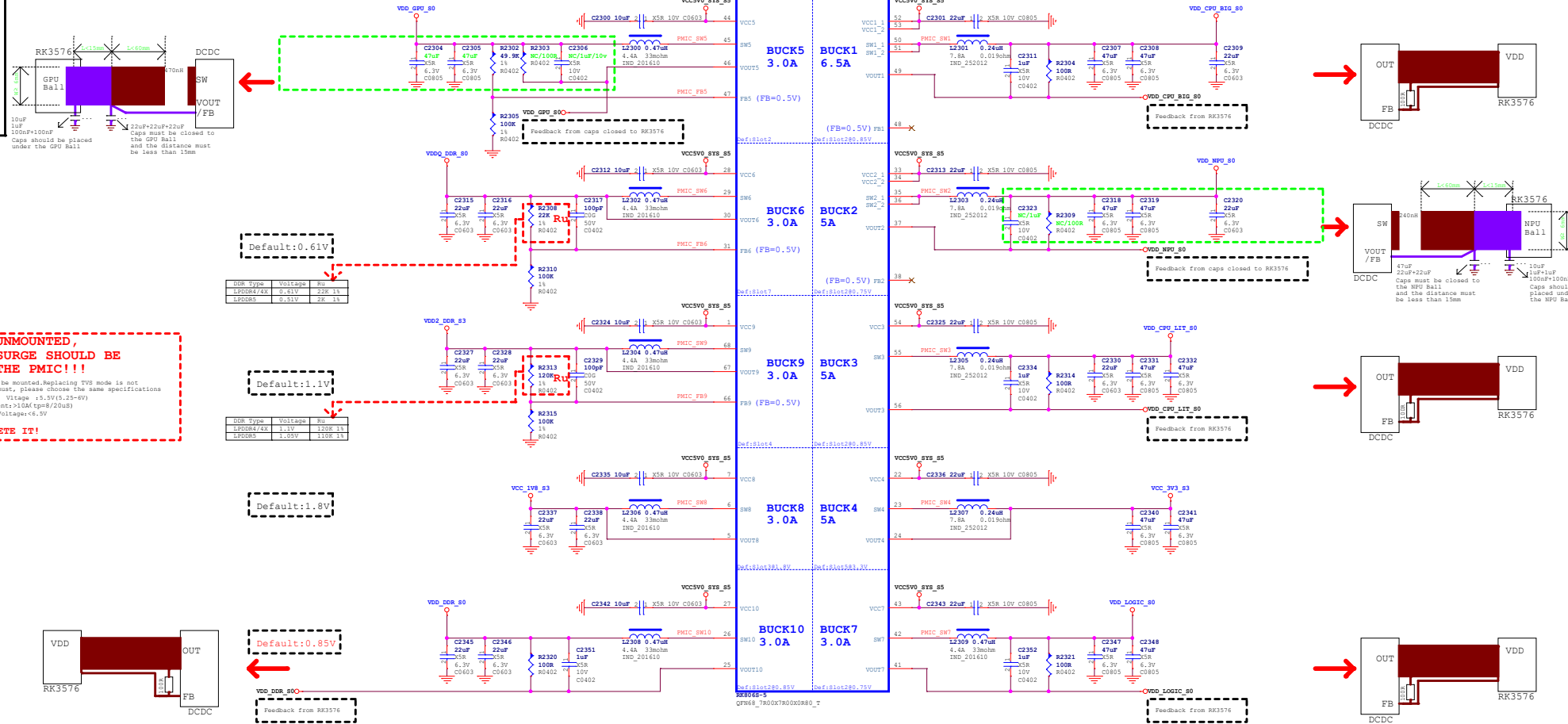
11] 12C1_SDA_M0_RK804 >>>
11] 12C1_SCL_M0_RK804 >>>

[11] PMIC_FWR_CTRL1 >>>
[11] PMIC_FWR_CTRL2 >>>

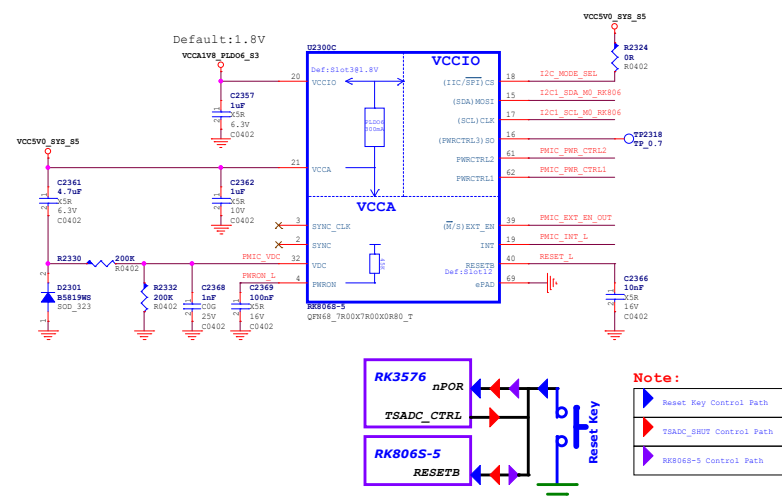
[11] PMIC_INT_1 <<<
[11,41] RESET_1 >>>

[22] PMIC_EXT_EN_OUT <<<
[41] PWRON_1 >>>

```



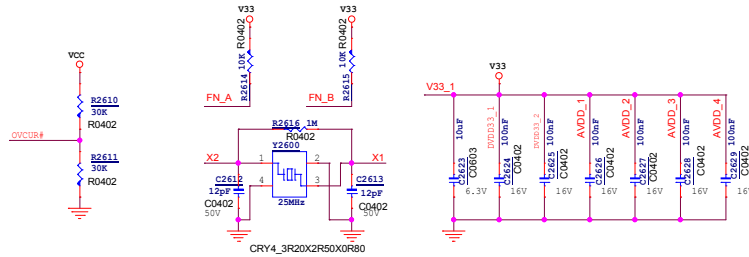
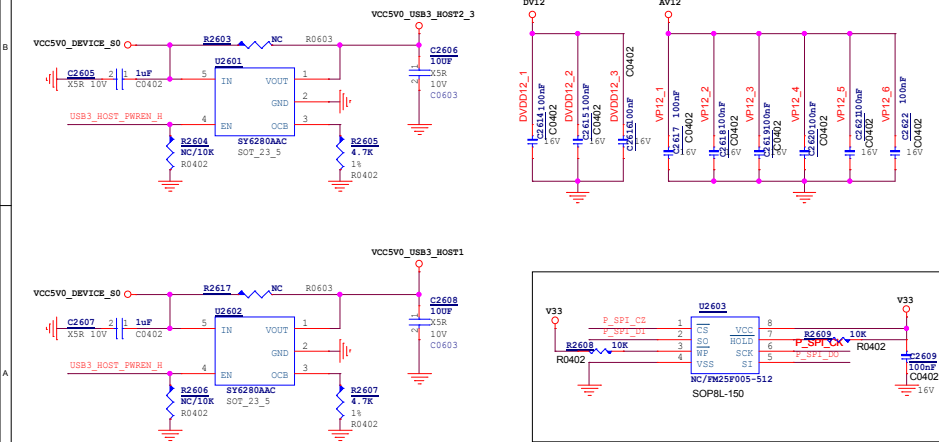
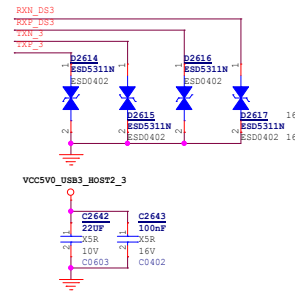
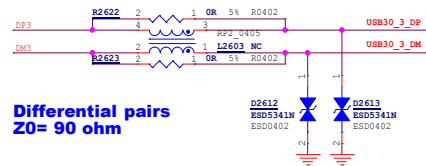
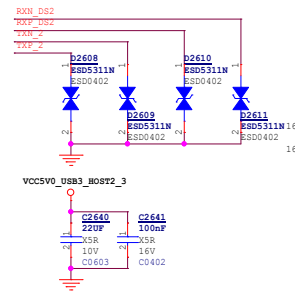
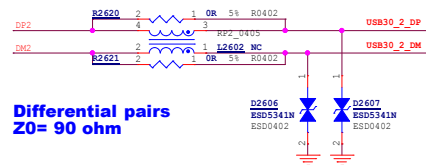
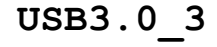
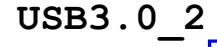
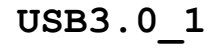
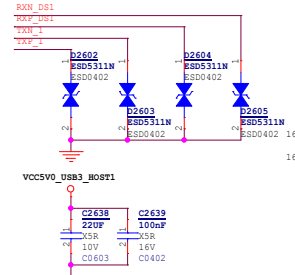
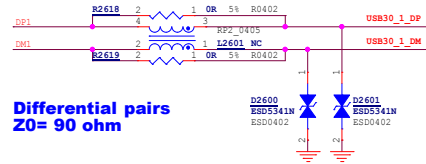
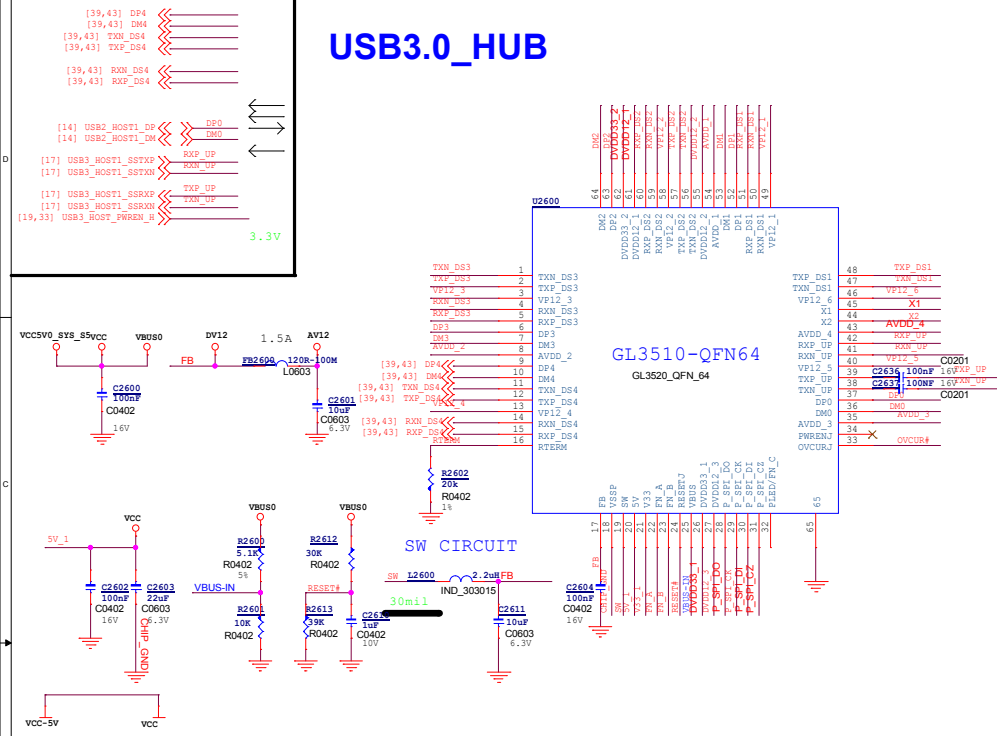
```
Note:
I2C Mode:CS(pin18) connected to VCCA(pin21);
SPI Mode(Def):CS(pin18) floating or connected to GND
```



The RK806 LDO power distribution of the reference schematics is only suitable for the interface used in the reference schematics. If other interface functions are to be added to the reference schematics, the RK806 LDO distribution needs to be re evaluated, otherwise the added

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Power-PMIC RK806S-5	K7 V1.1
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USB3.0_HUB



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eMMC FLASH

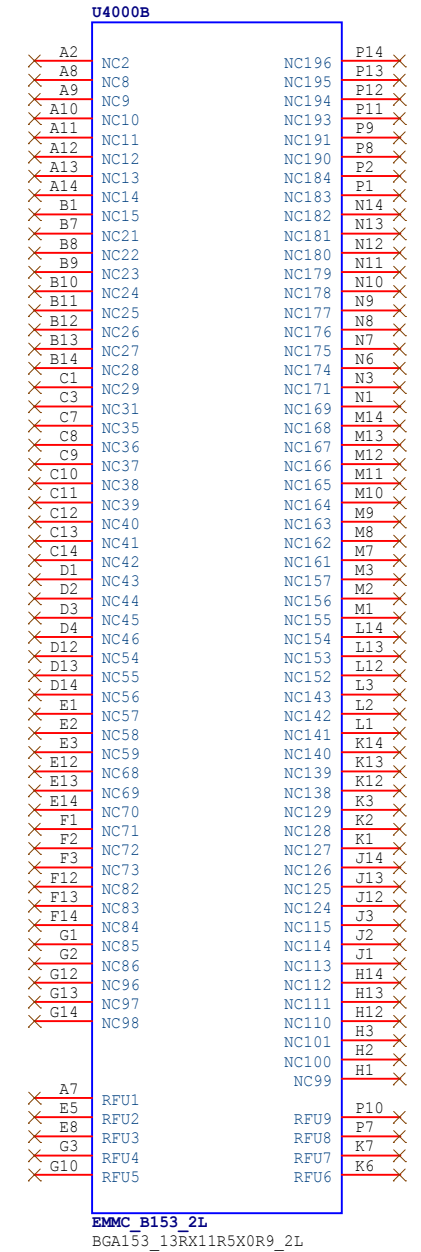
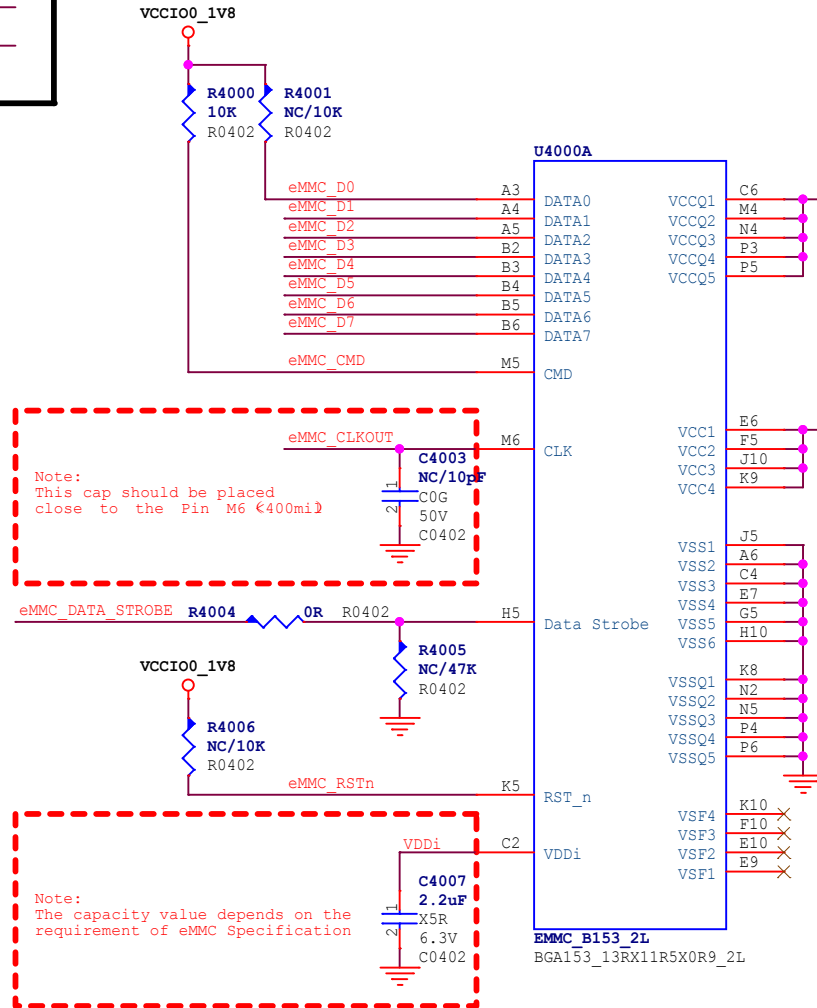
[13] eMMC_D0<<>>
[13] eMMC_D1<<>>
[13] eMMC_D2<<>>
[13] eMMC_D3<<>>
[13] eMMC_D4<<>>
[13] eMMC_D5<<>>
[13] eMMC_D6<<>>
[13] eMMC_D7<<>>

[13] eMMC_CMD<<>>

[13] eMMC_CLKOUT<<>>

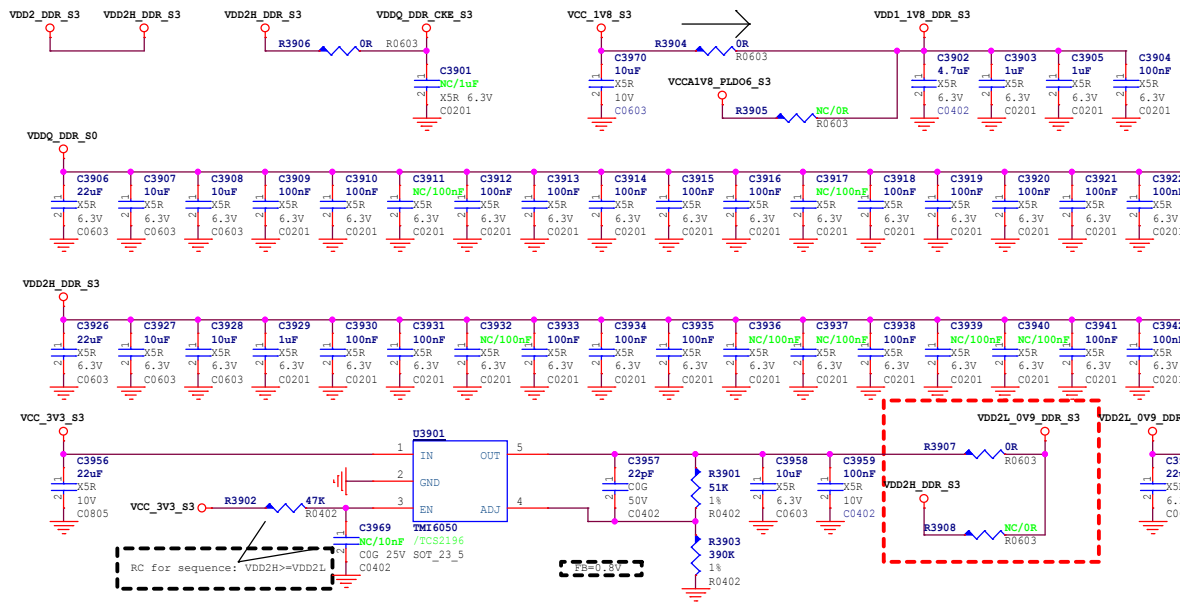
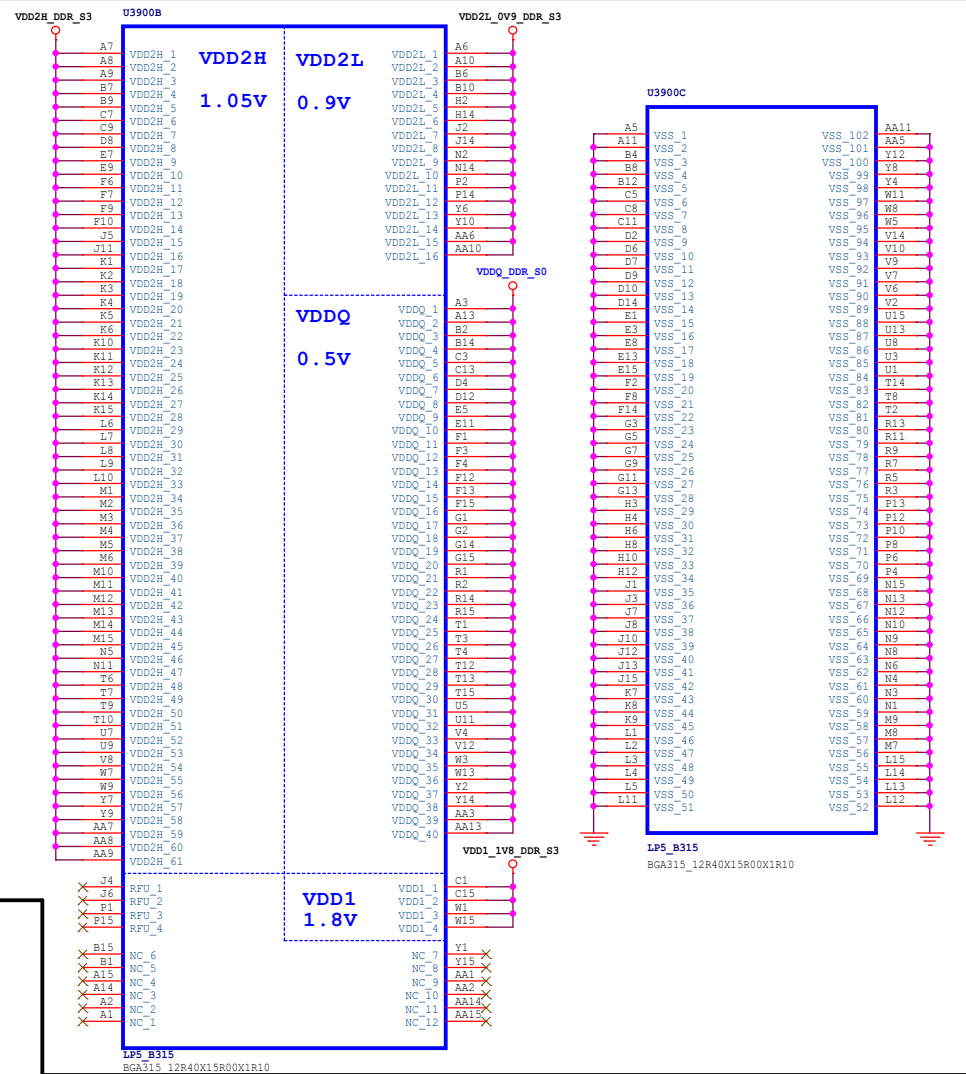
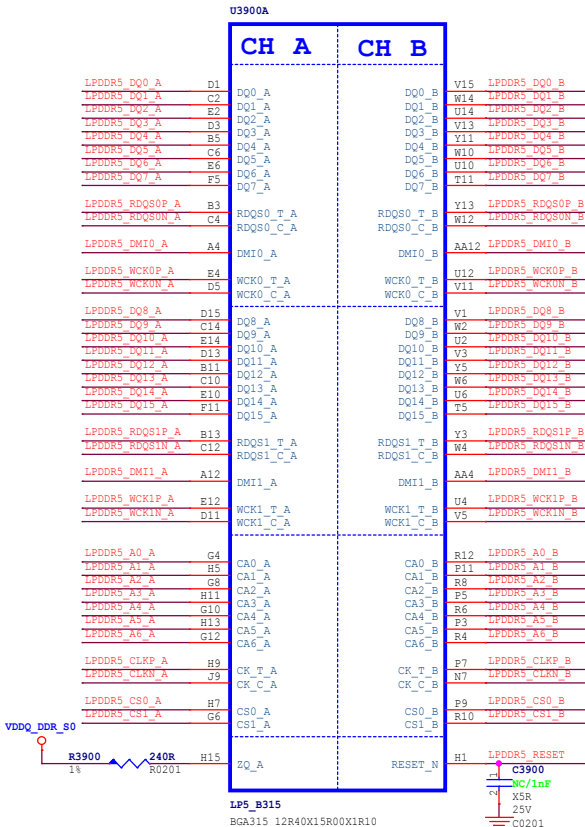
[13] eMMC_DATA_STROBE<<>>

[13] eMMC_RSTn<<>>



Oranthe		Document Title	
深圳市橙视科技发展有限公司		K7(RK3576+LPDDR4)	
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Flash-eMMC		K7_V1.1	
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LPDDR5



Note:

Power Sequence: VDD1>VDD2H>VDD2L>VDDQ-200mV

VDD1: 1.70V~-1.95V, Power for core1.

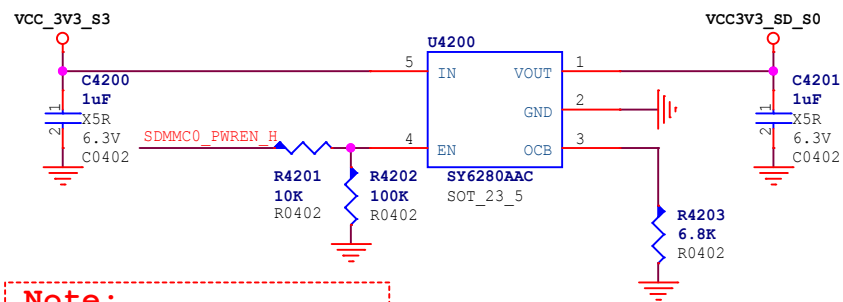
VDD2H: 1.01V~-1.12V, Power for core2 and Input Buffer.

VDD2L: 0.87V~-0.97V, Power for core2 and Input Buffer.

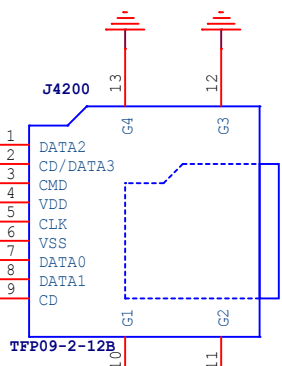
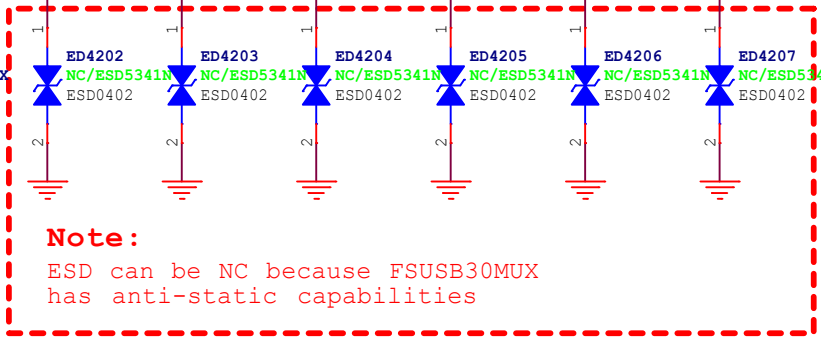
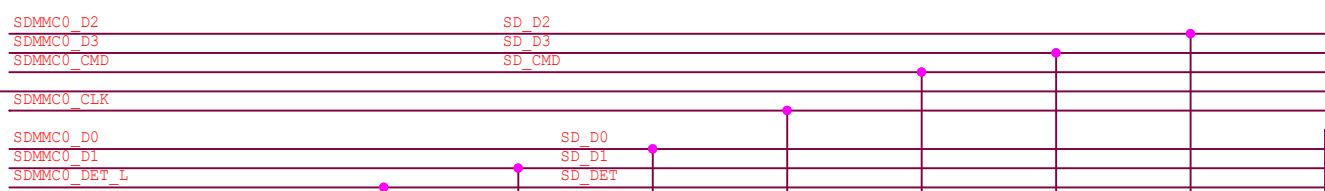
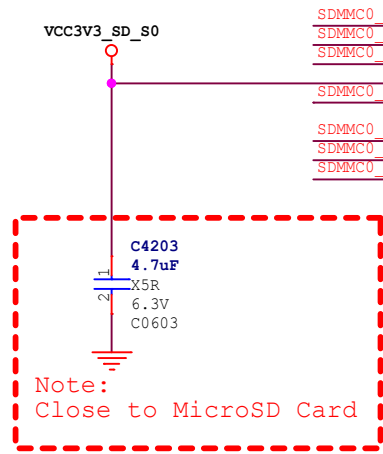
VDDQ: 0.47V~-0.57V, Power for I/O Buffer.

TF CARD

- [13] SDMMC0_D0
- [13] SDMMC0_D1
- [13] SDMMC0_D2
- [13] SDMMC0_D3
- [13] SDMMC0_CMD
- [13] SDMMC0_CLK
- [11] SDMMC0_DET_I
- [11] SDMMC0_PWREN_H



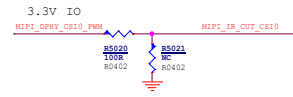
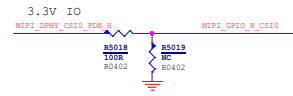
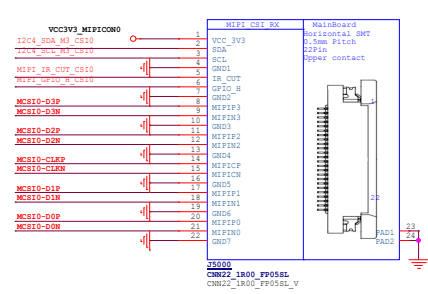
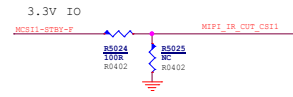
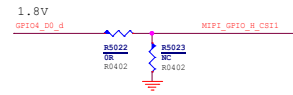
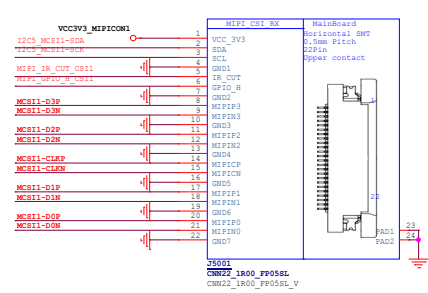
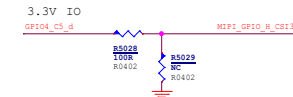
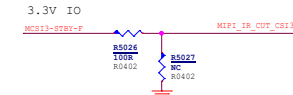
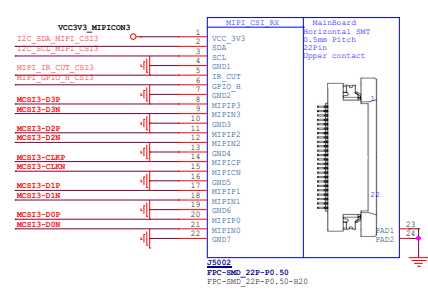
Note:
SDMMC_PWREN=H VCC3V3_SD(Default)
SDMMC_PWREN=L VCC3V3_SD=0V



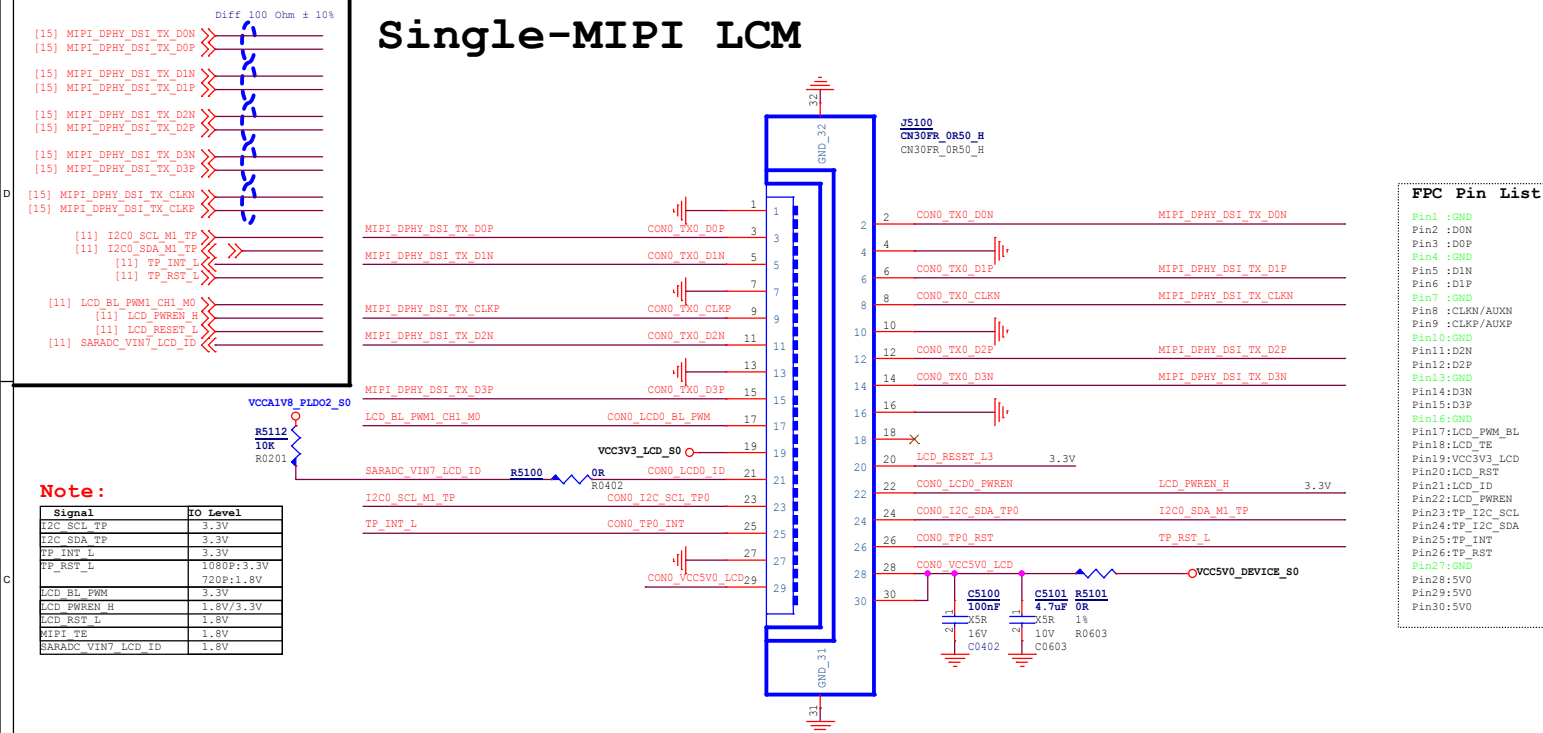
MicroSD Card

Note:
SDMMC_DET_L:
SDCARD PLUG: Pull-down to GND
SDCARD UNPLUG: Pull-up to PMUIO0_VCC1V8.

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深圳市橙视科技发展有限公司		K7(RK3576+LPDDR4)	
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Flash-MicroSD Card		K7_V1.1	
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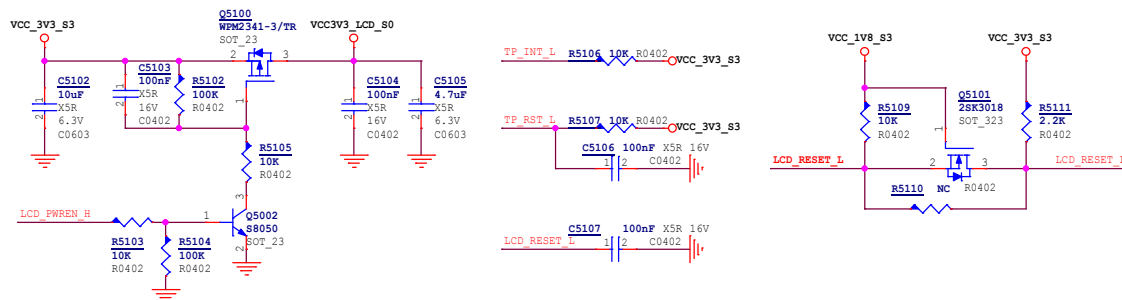
[illegible][illegible][illegible]

Single-MIPI LCM



Note:

Signal	IO Level
I2C_SCL_TP	3.3V
I2C_SDA_TP	3.3V
FP_INT_L	3.3V
FP_RST_L	1080P:3.3V 720P:1.8V
LCD_BL_PWM	3.3V
LCD_PWREN_H	1.8V/3.3V
LCD_RST_L	1.8V
MIP1_TE	1.8V
SARADC_VIN7_LCD_ID	1.8V



		Document Title	
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HDMI 2.1 Support video output up to 4Kx2K@120Hz

Note:

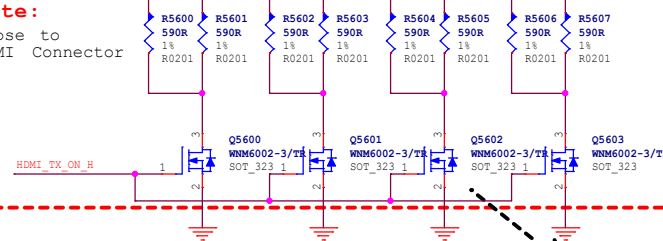
Close to HDMI Connector

HDMI_TX_D0P	C5600	1	2	220nF	C0201	X5R	10V
HDMI_TX_D0N	C5601	1	2	220nF	C0201	X5R	10V
HDMI_TX_D1P	C5602	1	2	220nF	C0201	X5R	10V
HDMI_TX_D1N	C5603	1	2	220nF	C0201	X5R	10V
HDMI_TX_D2P	C5604	1	2	220nF	C0201	X5R	10V
HDMI_TX_D2N	C5605	1	2	220nF	C0201	X5R	10V
HDMI_TX_D3P	C5606	1	2	220nF	C0201	X5R	10V
HDMI_TX_D3N	C5607	1	2	220nF	C0201	X5R	10V

Diff 100 Ohm ± 10%

Note:

Close to HDMI Connector



Rds=1.7ohm/2.6V
Coss=7.33pf

Note:

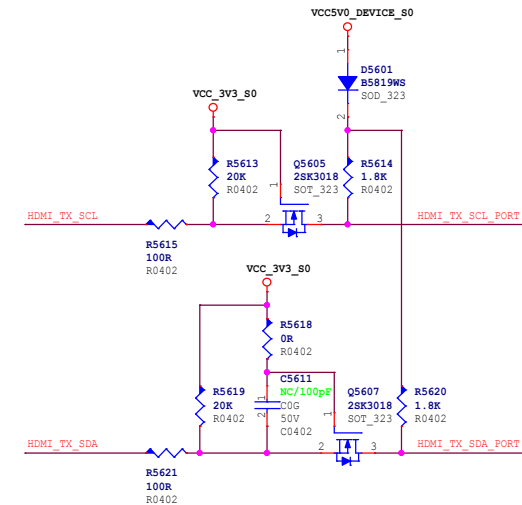
The HDMI2.1 trace length is less than 100mm.
The HDMI2.1 differential trace impedance is 100 OHM.

Note:

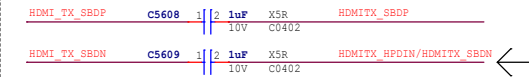
The controller only support AC coupled link. In order to backward compatibility, or to meet HDMI2.0 (1.4b) DC common mode spec and Voff, need do R based level-shift.

Switch on in HDMI2.0(TMDS) mode
Switch off in HDMI2.1(FRL) mode.

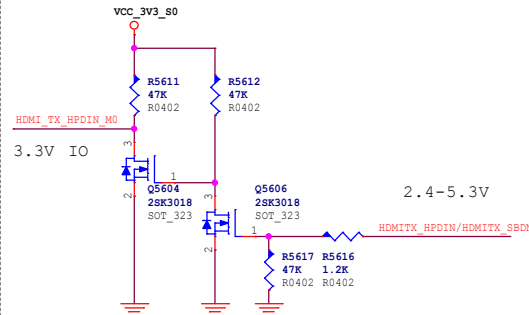
HDMI TX DDC



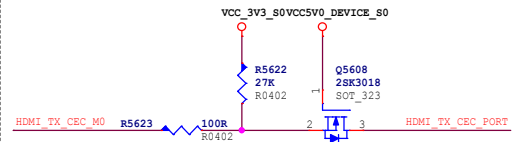
HDMI TX ARC



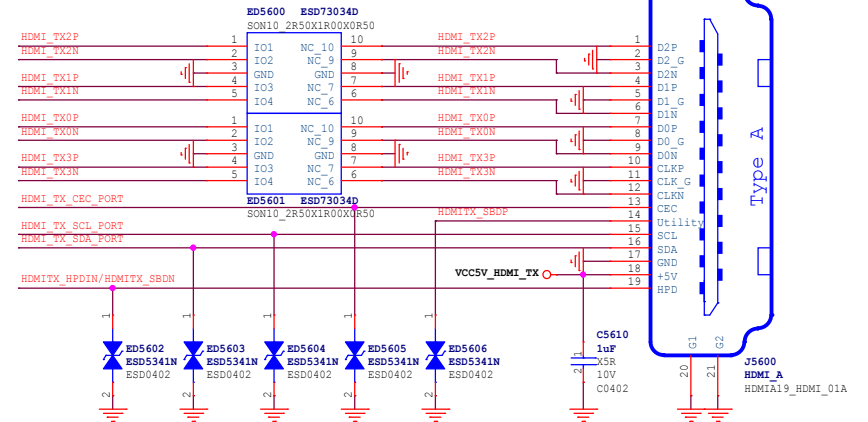
HDMI TX HPD



HDMI TX CEC

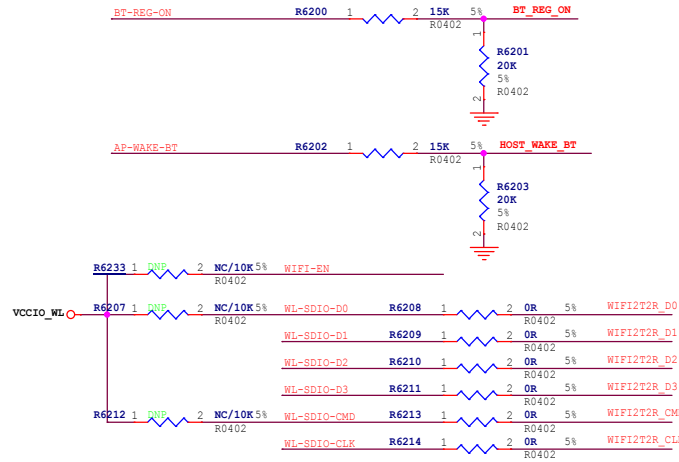


Cj<=0.2pF



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深圳市煜视科技发展有限公司		K7(RK3576+LPDDR4)	
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VO-HDMI TX		K7 V1.1	
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1.8V [18] SDMMC1_CLK_M0 WL-SDIO-CLK
[18] SDMMC1_CMD_M0 WL-SDIO-CMD
[18] SDMMC1_D0_M0 WL-SDIO-D0
[18] SDMMC1_D1_M0 WL-SDIO-D1
[18] SDMMC1_D2_M0 WL-SDIO-D2
[18] SDMMC1_D3_M0 WL-SDIO-D3
[18] UART4_RX_M1 BT-UART-RX
[18] UART4_TX_M1 BT-UART-TX
[18] UART4_CTS_M1 BT-UART-CTS
[18] UART4_RTS_M1 BT-UART-RTS
1.8V [18] BT_REG_ON_H BT-REG-ON
[18] WIFI_REG_ON_H WIFI-EN
[18] HOST_WAKE_BT_H AF-WAKE-BT
1.8V [11] BT_WAKE_HOST_H BT-WAKE-AP
[18] WIFI_WAKE_HOST_H WL-WAKE_HOST
[22] 32KOUT_RTC2CON R0402



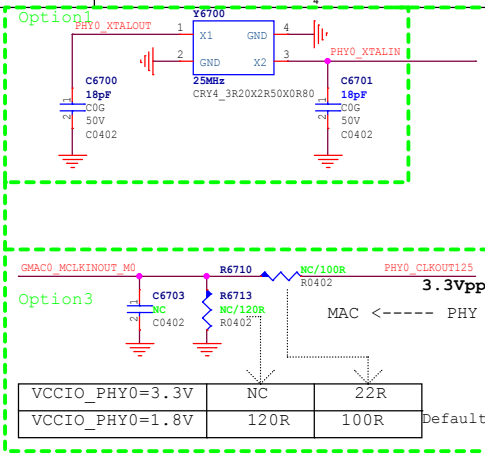

```
[19] GMAC0_TXD0_M0
[19] GMAC0_TXD1_M0
[19] GMAC0_TXD2_M0
[19] GMAC0_TXD3_M0
[19] GMAC0_TXCTL_M0
[19] GMAC0_TXCLK_M0
```

```
[19] GMACO_RXD0_M0 << _____
[19] GMACO_RXD1_M0 << _____
[19] GMACO_RXD2_M0 << _____
[19] GMACO_RXD3_M0 << _____
[19] GMACO_RXCTL_M0 << _____
[19] GMACO_RXCLK_M0 << _____

[19] GMACO_MCLKINOUT_M0 << _____
[19] GMACO_MDC_M0 << _____
[19] GMACO_MDIO_M0 << _____
[19] GMACO_RSTn_M0 << _____

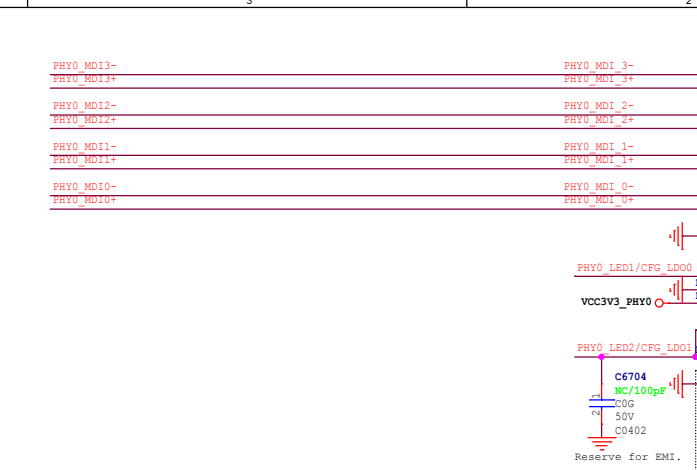
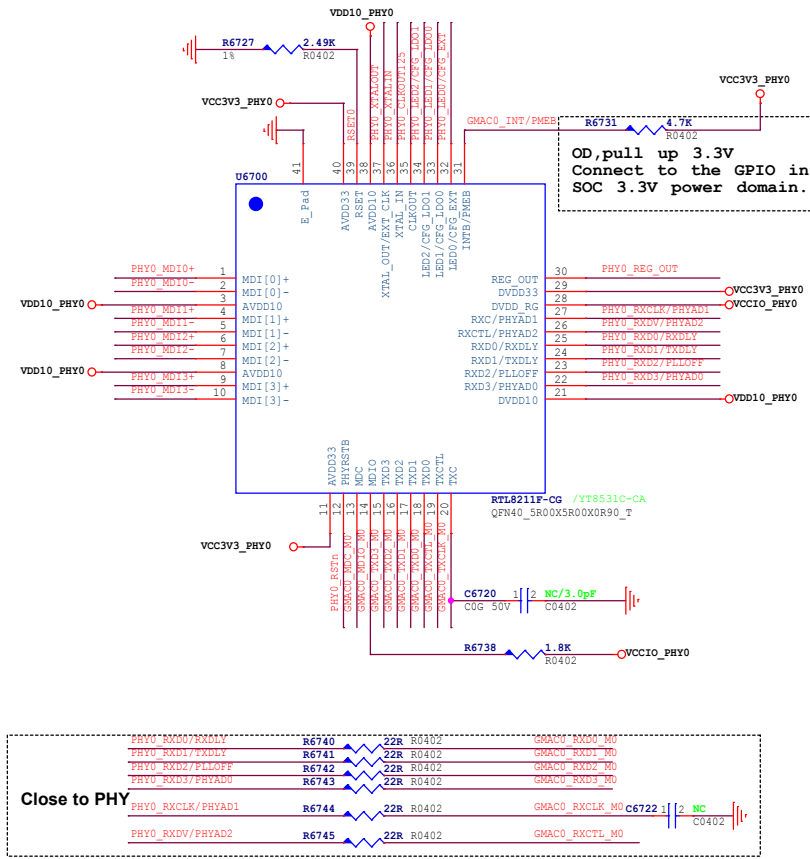
[19] GMACO_INT/PMBE >> GMACO_INT/PMBE
```

RK SOC clock mode recommended	
RK3576	mode1,mode2



VCCIO_PHY0=3.3V	NC	22R	Default
VCCIO_PHY0=1.8V	120R	100R	

Giga PHY clock mode selected			
Clock mode	Option1	Option2	Option3
mode1	No	Yes	No
mode2	Yes	No	No
mode3	Yes	No	Yes



The diagram shows three input lines on the left connected to three resistors (R6718, R6720, R6722) with a value of 4.7K and R0402. The other end of these resistors is connected to three pins: PHY0_LED0/CFG_EXT, PHY0_LED2/CFG_LD01, and PHY0_LED1/CFG_LD00. Additionally, there is a connection from a pin labeled NC/4.7K through a 4.7K resistor and R0402 to a pin labeled VCC3V3_PHY0.

VCC_PHY0_IO Voltage Config



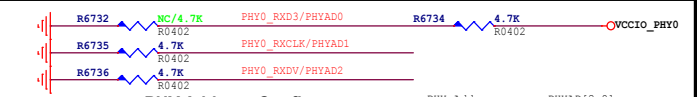
Pull-up to disable PLL @ ALDPS mode(Low power mode)



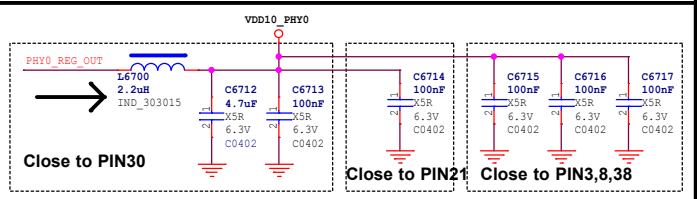
Pull-up for additional 2ns delay to RXC for data latching



Pull-up for additional 2ns delay to TXC for data latching

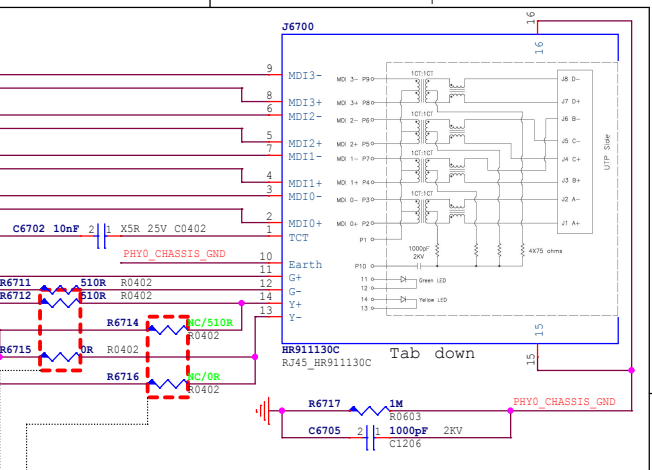
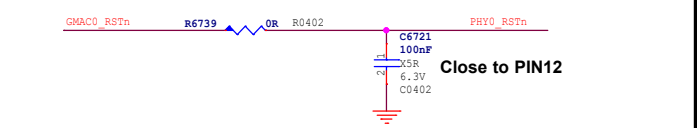


PHY Address	PHYAD[2:0]
1 (default)	3'b001

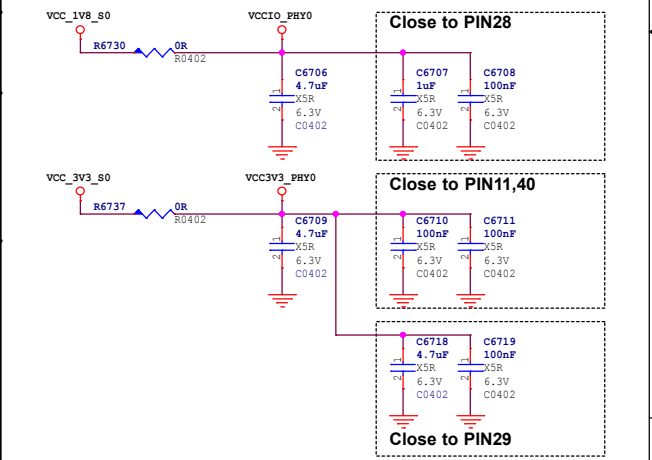


Close to PIN30

PHYRSTB is 3.3V IO



RGMII Power Source	CFG_EXT	CFG_LDO[1:0]	PHY0_LED2
External 3.3V	1'b1	2'b00	String S10R to GND
External 1.8V	1'b1	2'b10	String S10R to VCC3V3_PHY
Internal 1.8V (default)	1'b0	2'b10	String S10R to VCC3V3_PHY



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Ethernet-GEPHY		K7_V1.1	
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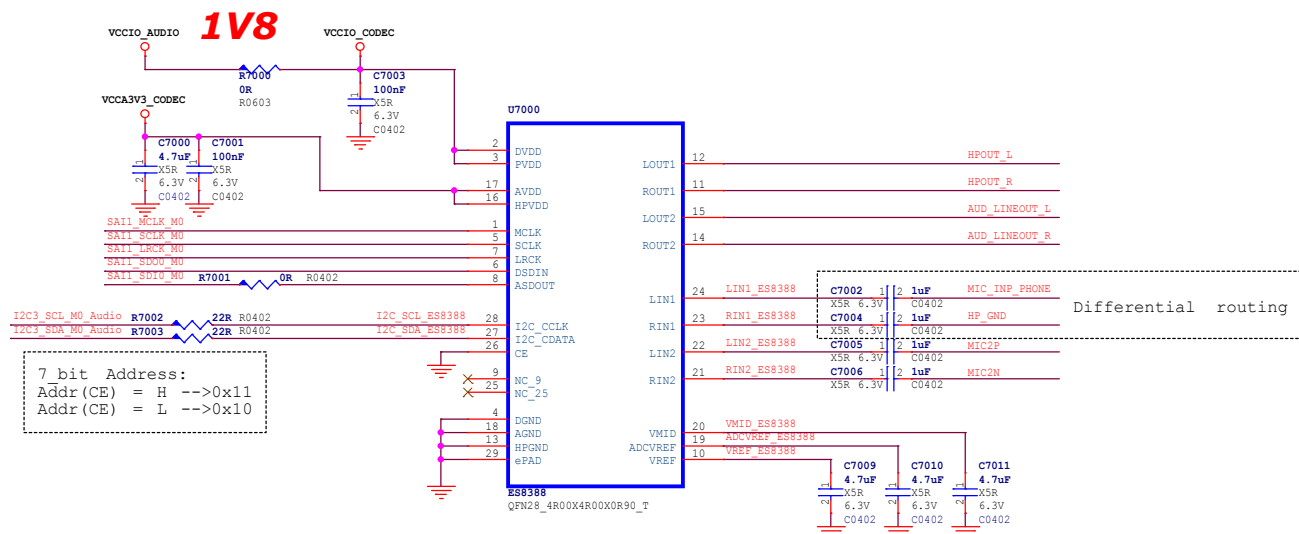
CODEC

[18] SA11_MCLK_M0 >>>
[18] SA11_SCLK_M0 >>>
[18] SA11_LCLK_M0 >>>
[18] SA11_SDO0_M0 >>>
[18] SA11_SDI0_M0 >>>

[38] AUD_LINEOUT_L >>>
[38] AUD_LINEOUT_R >>>
[11] HP_DET_L >>>

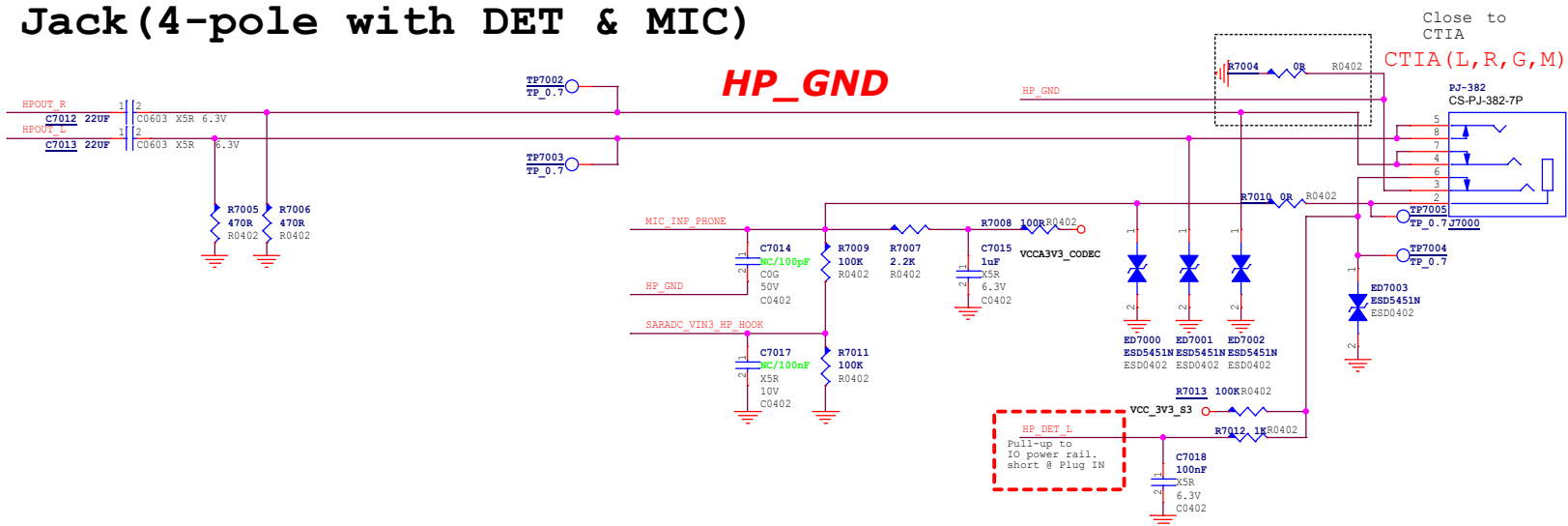
[11] SARADC_VIN3_HP_HOOK <<<

[18] I2C3_SCL_M0_Audio >>>
[18] I2C3_SDA_M0_Audio >>>

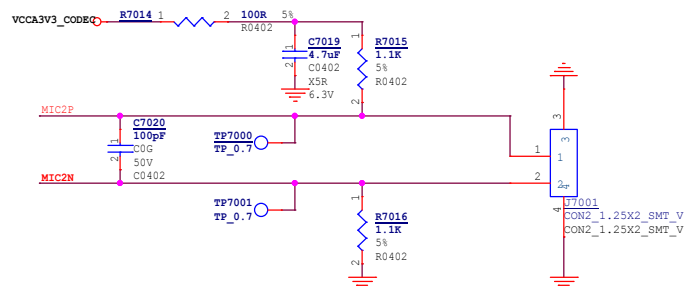


The chip can adopt 1.8V power supply to reduce power consumption

Headphone Jack(4-pole with DET & MIC)



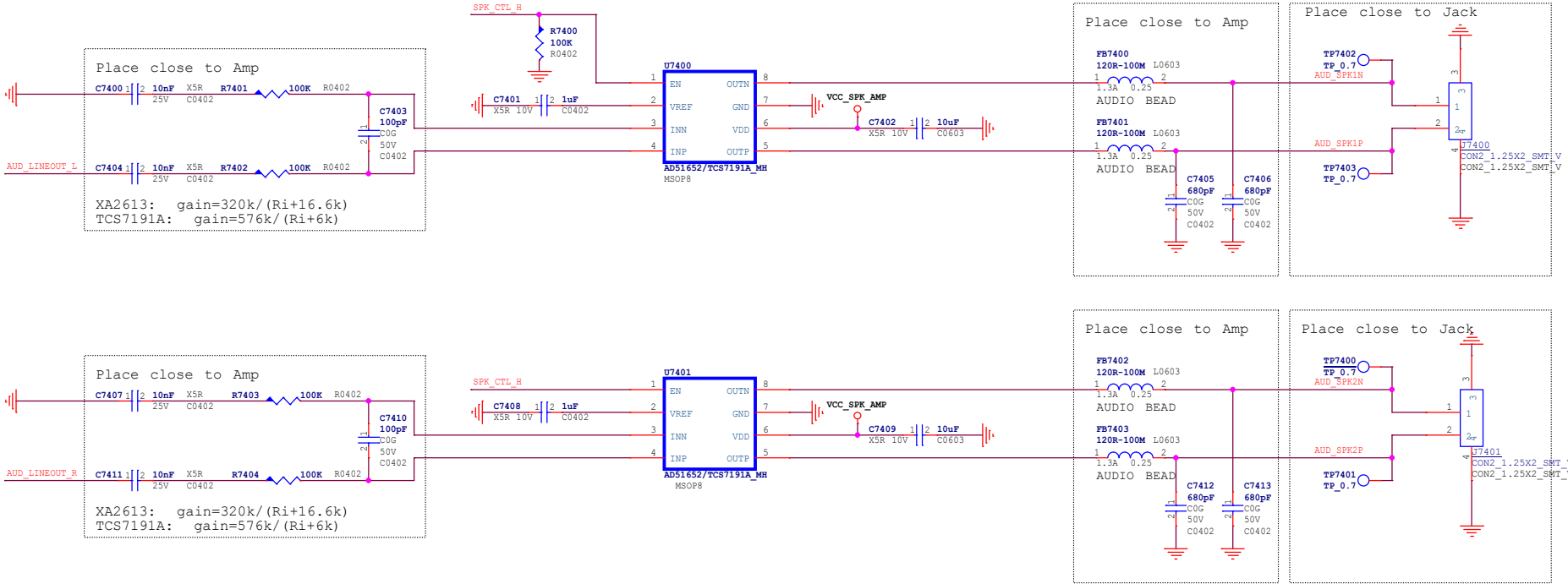
MIC



Shenzhen 深圳市橙视科技发展有限公司		Document Title K7(RK3576+LPDDR4)	
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Speaker Output

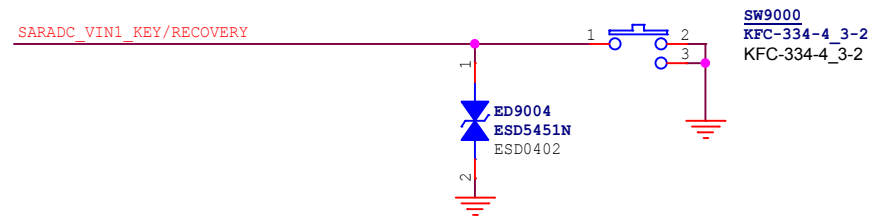
[37] AUD_LINEOUT_L
[37] AUD_LINEOUT_R
[19] SPK_CTL_H



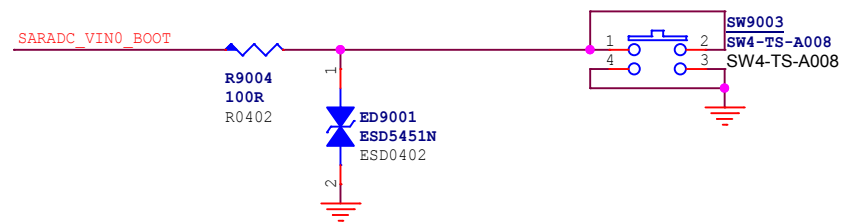
Qianxin		Document Title	
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[11] SARADC_VIN1_KEY/RECOVERY
[11,21] RESET_L
[21] PWRON_L
[11] SARADC_VIN0_BOOT

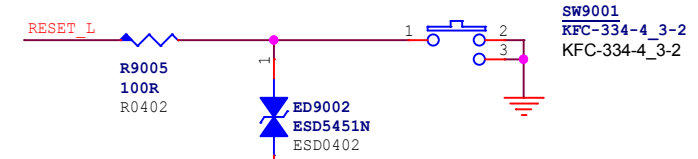
RECOVERY



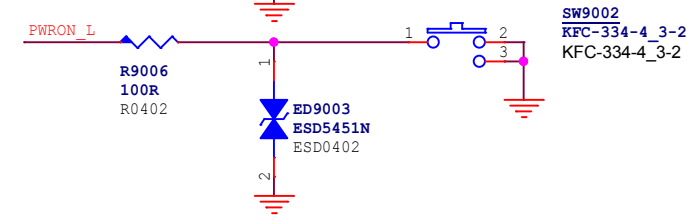
MASKROM Key



RESET



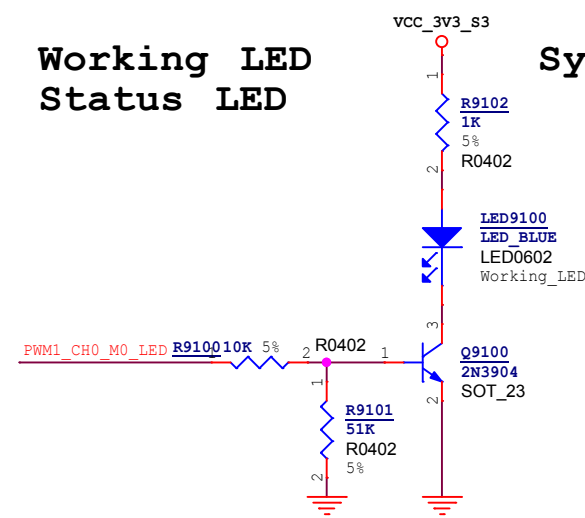
PWRON



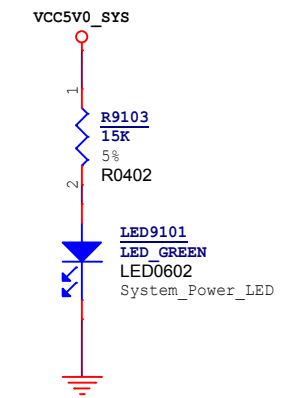
Oranth		Document Title	
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Page Title		PCB NO.	
Key-PowerON/Reset		K7_V1.1	
Page Size	Sch Designer	Pcb Designer	Project Stage
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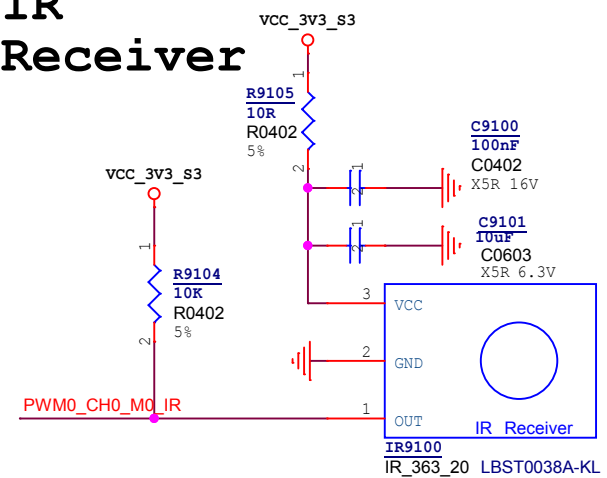
Working LED Status LED



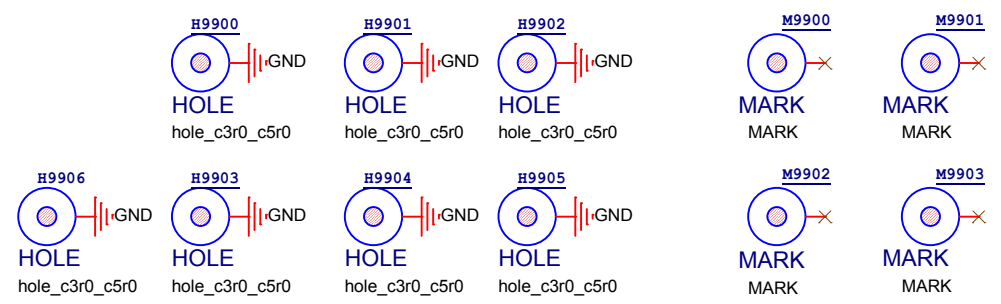
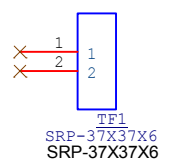
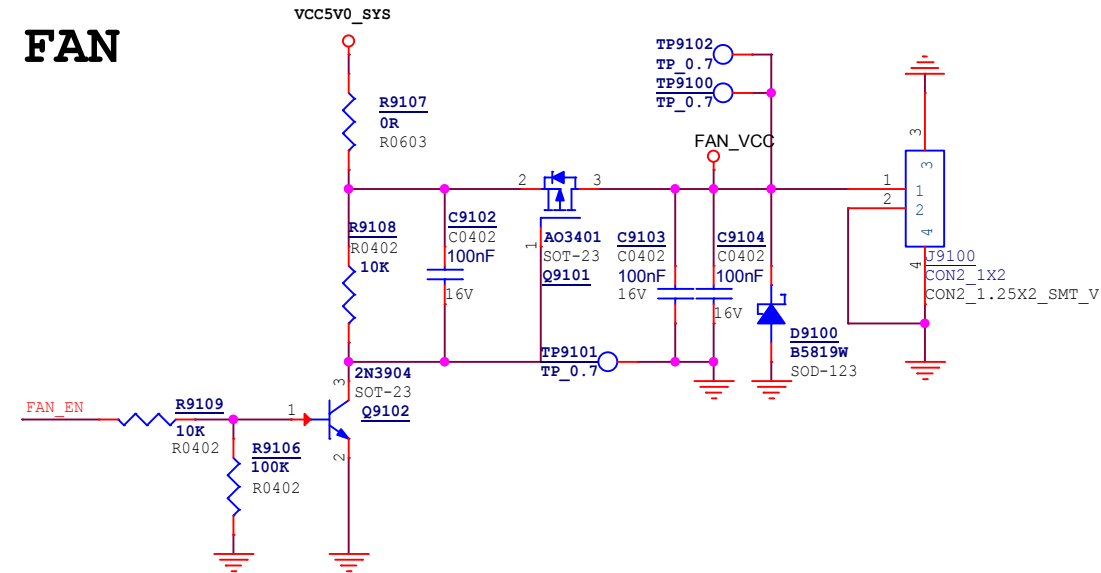
System_Power_LED



IR Receiver



FAN



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```

┌──────────────────┴──────────────────┐
└──UART0_RX_M0_DEBUG [11]
└──UART0_TX_M0_DEBUG [11]

```

```
GPIO1_D2_d  >>> I3S0_SCL_M1 [18]
GPIO1_D3_d  >>> I3S0_SDA_M1 [18]
```

The diagram shows two horizontal lines representing the I2C7_SCL_M1 and I2C7_SDA_M1 pins of the I2C7 module. These lines are connected to the I2C7_SCL_M1 and I2C7_SDA_M1 pins of the I2C7 module, which are labeled [19].

```
GPIO1_D0_d  >>> UART10_TX_M1[18]
GPIO1_D1_d  >>> UART10_RX_M1 [18]
```

SARADC_VIN4 SARADC_VIN4 [11]
SARADC_VIN5 SARADC_VIN5 [11]
SARADC_VIN6 SARADC_VIN6 [11]

```
GPIO4_A4_d      >>> GPIO4_A4_d [18] 1v8
GPIO4_A6_d      >>> GPIO4_A6_d [18] 1v8
```

GPIO4_B0_d	PDM1_CLK1_M1		PDM1_CLK1_M1 [18]	1v8
GPIO4_B1_d	PDM1_SDI2_M1		PDM1_SDI2_M1 [18]	1v8
GPIO4_B2_d	PDM1_SDI1_M1		PDM1_SDI1_M1 [18]	1v8
			PDM1_SDI1_M1 [18]	1v8

GPIO3_B0_d << GPIO3_B0_d [19] 1v8

GPIO3_D4_d >>> GPIO3_D4_d [19] **1V8**
GPIO3_D5_d >>> GPIO3_D5_d [19] **1V8**

GPIO2_D7_d >> GPIO2_D7_d [19] 3V3
GPIO2_D6_d >> GPIO2_D6_d [19] 3V3

```
GPIO3_C5_d  >>> MIPI_DPHY_CSI2_PDN_H [19]
GPIO3_C6_d  >>> MIPI_DPHY_CSI0_PWREN_H [19]
```

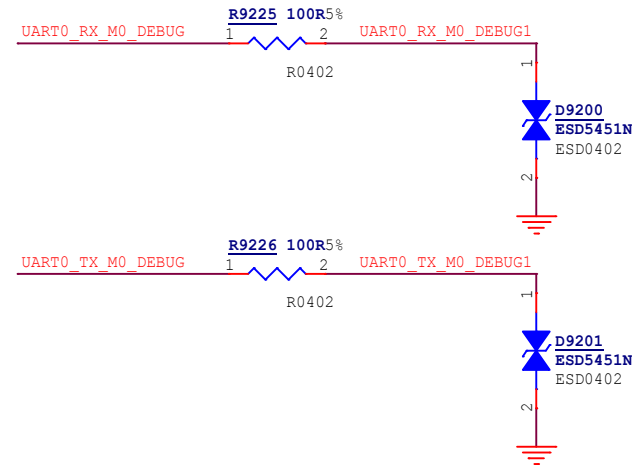
```

GPIO4_C6_d
GPIO4_C7_d
GPIO4_C6_d
GPIO4_C7_d

```

GPI00_C3_d >> GPI00_C3_d [11] 3v3

GPIO0_A5_d [11] 1v8

[illegible]

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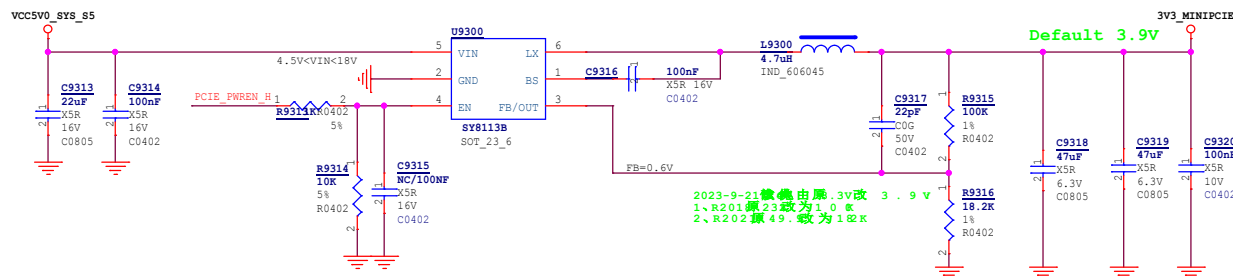
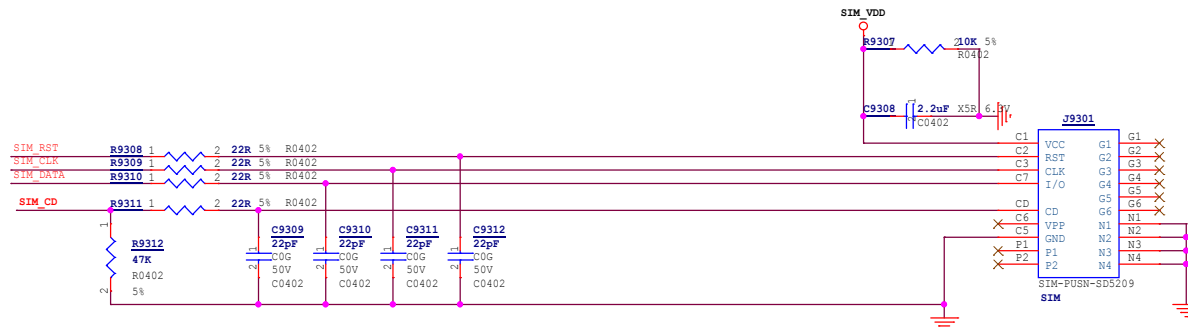

```

[11] PCIE_PWREN_H << PCIE_PWREN_H
      [43] DM4 << >>
      [43] DP4 << >>
[39,43] TXN_DS4 <<
[39,43] TXP_DS4 <<
      [39,43] RXN_DS4 <<
      [39,43] RXP_DS4 <<

```

The diagram illustrates the internal wiring of the J9300 Mini-PCIe module. Key components and connections include:

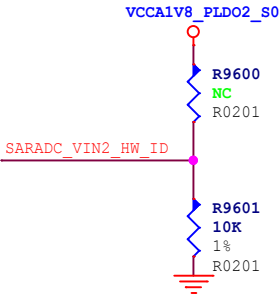
- Pin Headers:**
 - [39,43] RXN_DS4** and **[39,43] RXF_DS4** are connected to pins 1 and 2 of the J9302 header.
 - [39,43] TXN_DS4** and **[39,43] TXP_DS4** are connected to pins 3 and 4 of the J9301 header.
- Internal Components:**
 - J9300:** The main Mini-PCIe module, labeled CS-C-91304A-52H.
 - J9301:** A 16V 100nF capacitor connected to pin 35 (GND5) and pin 36 (USB_DM).
 - J9302:** A 10V C0402 capacitor connected to pin 41 (VCC_3V3_1) and pin 42 (VCC_3V3_2).
 - J9303:** A 10V C0402 capacitor connected to pin 43 (GND7) and pin 44 (PCW_CLK).
 - J9304:** A 10V X5R capacitor connected to pin 45 (PCW_OUT) and pin 46 (PCW_IN).
 - J9305:** A 10V C0402 capacitor connected to pin 47 (PCW_SYNC) and pin 48 (GND8).
 - J9306:** A 10V C0402 capacitor connected to pin 49 (VCC_3V3_1) and pin 50 (VCC_3V3_2).
- External Connections:**
 - 3V3_MINIPICIE:** Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52.
 - SIM_VDD:** Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52.
 - SIM_DATA:** Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52.
 - SIM_CLK:** Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52.
 - SIM_RST:** Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52.
 - SIM_CD:** Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52.



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[11] SARADC_VIN2_HW_ID<<<

HW_ID



Config Table for SARADC_VIN2_HW_ID				
Item	Rup	Rdown	ADC Value	VERSION
HW_ID1	NC	10K	0	RK3576 EVB1 V12
HW_ID2	10K	1.13K	416	RESERVE
HW_ID3	10K	2.49K	816	RESERVE
HW_ID4	10K	4.3K	1231	RESERVE
HW_ID5	10K	6.8K	1658	RESERVE
HW_ID6	10K	10K	2048	RESERVE
HW_ID7	10K	14.7K	2437	RESERVE
HW_ID8	10K	23.2K	2862	RESERVE
HW_ID9	10K	40.2K	3279	RESERVE
HW_ID10	10K	88.7K	3680	RESERVE
HW_ID11	10K	NC	4095	RESERVE

SARADC PIN